

Space-Time Algebra (STA)

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Abstract—Space-time algebra (STA) is an associative algebra that combines scalars and vectors into a single object called a paravector. In this form, the algebra is adapted to rotations, Lorentz boosts, space-time kinematics, and relativistic field theory. This manuscript develops a common STA notation for proper velocity and momentum, Maxwell electrodynamics, and Klein–Gordon, Dirac, and Pauli-type wave equations.

Index Terms—space-time algebra, geometric algebra, Lorentz transformations, Maxwell electrodynamics, Dirac equation, Pauli equation.

I. INTRODUCTION

Space-time algebra extends the familiar scalar and vector calculus of three-dimensional space into a unified algebra adapted to rotations, Lorentz transformations, and relativistic field equations. In this manuscript the basic object is the paravector, a real scalar plus a real vector, and the central claim is organizational as much as technical: the same algebraic product can be carried from elementary vector identities to kinematics, electromagnetism, and relativistic wave equations without changing mathematical language from one topic to the next.

The exposition is arranged so that each layer depends only on material already established. Section II fixes the algebra itself; Sections III and IV reinterpret that algebra as space-time geometry and covariant motion; Section V develops the electromagnetic sector; and Section VI uses the same machinery for Klein–Gordon, Dirac, and Pauli-type equations. A compact formula compendium is deferred to Appendix A, while representation theory, verification tables, historical notes, and cautions are kept in later appendices so they do not interrupt the main line of argument.

This perspective is consistent with the modern STA and paravector literature of Hestenes, Baylis, and Doran–Lasenby [9–11], but the notation is kept deliberately lean. Each major section is developed in two passes: a compact forward exposition that introduces only the formulas needed for the next step, followed by a local supporting development that records intermediate identities, tables, and special cases near the formulas they support. Readers interested in the shortest route through the argument can move from one numbered section to the next and return to the supporting derivations afterward. Symbolic checks are collected in Appendix F, and bold red notes mark formulas that require qualification or correction.

Lower-case Latin letters denote real scalars unless stated otherwise, bold letters denote real three-vectors, capital letters denote paravectors or operators according to context, and complexification is carried by the pseudoscalar i . Natural units $\hbar = c = 1$ and metric signature $(+, -, -, -)$ are used unless stated otherwise.

A. Reading Guide

1. Section II defines the sigma basis, involutions, paravector products, quadratic forms, exponentials, and projectors.
2. Sections III and IV recast those algebraic objects as interval geometry, proper kinematics, rotors, and Lorentz boosts.
3. Section V introduces the paravector differential operator, fields, potentials, Maxwell theory, gauge structure, and the Lorentz force.
4. Section VI develops the relativistic wave-equation sector, from Klein–Gordon factorization to the Dirac block system and the Pauli limit.
5. Appendix A collects the principal formulas in one place after the narrative has been established.
6. Appendices B–E gather supplementary representation links, formula-status tables, historical context, and explicit cautions, while Appendix F records the independent symbolic checks.

II. ALGEBRAIC FOUNDATIONS

The formula compendium in Appendix A is deliberately complex-first. The exposition now resets to the simplest ingredients: basis rules, the sigma-conjugate and reverse operators, vector products, paravectors, and then the elementary functional calculus built from them.

Basis and pseudoscalar

The algebra starts with three generators σ_n for $n \in \{1, 2, 3\}$. These are basis algebra elements rather than scalars, and they behave like an orthonormal spatial basis whose multiplication carries more information than an ordinary dot product. From these generators one obtains vectors, bivectors, and the pseudoscalar i , which plays the role of the imaginary unit while remaining internal to the algebra.

The basis relations are

$$\begin{aligned} \sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\ i &:= \sigma_1 \sigma_2 \sigma_3, & i^2 &= -1. \end{aligned} \quad (1)$$

Using δ_{nm} and ϵ_{nmk} for indices $n, m, k \in \{1, 2, 3\}$,

$$\sigma_n \sigma_m = \delta_{nm} + \epsilon_{nmk} i \sigma_k. \quad (2)$$

The Sigma-Conjugate, Reverse, and Combined Operators

The two basic operators are introduced here on their own because they are used repeatedly in every later section. Each is an involution, and the combined operator built from them is fixed at the same point by the direct rules

$$(\cdot)^* := \text{sigma conjugate; flip each } \sigma_n, \quad (3)$$

$$(\cdot)^R := \text{reverse the generator order}, \quad (4)$$

$$(\cdot)^{*R} := \text{sigma conjugate, then reverse}. \quad (5)$$

Equation (3) defines the sigma-conjugate operator, Eq. (4) defines the reverse operator, and Eq. (5) defines the combined sigma-conjugate/reverse operator. The sigma-conjugate operator $(\cdot)^*$ therefore changes the sign of each basis generator while leaving product order unchanged, the reverse operator $(\cdot)^R$ leaves each σ_n fixed while reversing the order of generator products, and the combined operator $(\cdot)^{*R}$ performs both actions in succession.

Because sign-flip and order-reversal are independent, the combined operator may be applied in either order. In operator form,

$$\begin{aligned} (XY)^* &= X^* Y^*, & (XY)^R &= Y^R X^R, \\ X^{*R} &:= (X^R)^* = (X^*)^R. \end{aligned} \quad (6)$$

Applied to the generators and the pseudoscalar, these rules reduce to

$$\sigma_n^* = -\sigma_n, \quad \sigma_n^R = \sigma_n, \quad \sigma_n^{*R} = -\sigma_n, \quad (7)$$

$$i^* = -i, \quad i^R = -i, \quad i^{*R} = i. \quad (8)$$

These agree with the standard geometric-algebra involution structure described in the STA literature [9, 11]; throughout this manuscript they are referred to as the sigma-conjugate operator,

the reverse operator, and the combined sigma-conjugate/reverse operator. These operators will be used constantly, so it is worth emphasizing their distinct roles. The sigma-conjugate operator flips vector sign, the reverse operator reverses multiplication order, and the combined operation is the one that later produces the Minkowski-type quadratic form. Introducing them at the algebraic stage prevents the interval and operator formulas from seeming arbitrary when they appear later. The homogeneous sigma grades and the signs acquired under these three involutions are summarized here before the manuscript moves on to products and paravectors. These tables summarize the homogeneous sectors generated by σ_n and the way the sigma-conjugate, reverse, and combined operators act on each grade.

Table 1. Sigma grades and representative basis elements.

grade	sector	representative elements
0	scalar	1
1	vector	$\sigma_1, \sigma_2, \sigma_3$
2	bivector	$i\sigma = \{\sigma_2\sigma_3, \sigma_3\sigma_1, \sigma_1\sigma_2\}$
3	pseudo-scalar	$i = \sigma_1\sigma_2\sigma_3$

Table 2. Signs of the sigma-conjugate, reverse, and combined operators on homogeneous grades.

Grade	sigma conjugate ($\cdot$) [*]	reverse operator ($\cdot$) ^R	combined ($\cdot$) ^{*R}
scalar	+	+	+
vector	−	+	−
bivector	+	−	−
pseudo-scalar	−	−	+

Vector product identity and complex paravectors

Collect the three generators into the sigma-vector

$$\sigma := \{\sigma_1, \sigma_2, \sigma_3\}, \quad (9)$$

which is a vector of algebraic objects rather than a vector of scalar components. With that notation fixed, ordinary three-vectors act on the algebra by contraction against σ .

For real vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$,

$$(\mathbf{A} \cdot \sigma)(\mathbf{B} \cdot \sigma) = \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \sigma. \quad (10)$$

This real-vector product is the first structural identity because it makes the scalar/vector split transparent before any complexification is introduced. The fully complex product then follows as the master formula from which the simpler cases are recovered by restriction.

For real scalars $a, b \in \mathbb{R}$ and real vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, the corresponding complex scalar $S = a + ib \in \mathbb{C}$, complex vector $\mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3$, real paravectors $X, Y \in \mathbb{R} \oplus \mathbb{R}^3$, and complex paravector $Z \in \mathbb{C} \oplus \mathbb{C}^3$ are related by

$$Z = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \sigma = S + \mathbf{V} \cdot \sigma = X + iY, \quad (11)$$

with

$$\begin{aligned} S &= a + ib \in \mathbb{C}, & \mathbf{V} &= \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3, \\ X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, & Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3. \end{aligned} \quad (12)$$

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$ with $S, S' \in \mathbb{C}$ and $\mathbf{V}, \mathbf{V}' \in \mathbb{C}^3$,

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + i(\mathbf{V} \times \mathbf{V}')) \cdot \boldsymbol{\sigma}. \quad (13)$$

Equations (11) and (13) fix the generic complex paravector and its master multiplication rule: pure scalar, real vector, real paravector, and mixed products all follow by setting the unused parts to zero.

Declared types

Object	Type
$a, b \in \mathbb{R}$	real scalars
$\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$	real vectors
$S = a + ib \in \mathbb{C}$	complex scalar
$\mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3$	complex vector
$X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$	real paravector
$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3$	complex paravector

This table is archetypal rather than global. No letter is permanently tied to one type throughout the manuscript; instead, each symbol is assigned locally when it first appears in a new context. The basic ambient spaces are $\mathbb{R}, \mathbb{C}, \mathbb{R}^3, \mathbb{C}^3$, and the direct-sum paravector spaces $\mathbb{R} \oplus \mathbb{R}^3$ and $\mathbb{C} \oplus \mathbb{C}^3$.

The few objects that do not belong to one of those six spaces are signaled explicitly where they arise: the basis elements σ_n are algebra generators, the differential symbols $\partial, E, \mathbf{P}$, and Ω are operators, and the Dirac map $W(Z)$ together with $\mathbf{I}$ lives in a 2×2 block-matrix representation built from paravector entries. No additional scalar fields beyond $\mathbb{R}$ and $\mathbb{C}$ are used.

Two central quadratic forms are

$$\|Z\|^2 := \text{scalar}(ZZ^R) = |S|^2 + \|\mathbf{V}\|^2, \quad (14)$$

$$\langle\langle Z \rangle\rangle^2 := ZZ^{*R} = S^2 - \mathbf{V} \cdot \mathbf{V}. \quad (15)$$

Equation (15) is the paravector form later used for space-time intervals [10, 11]. The distinction between the Euclidean norm (14) and the Minkowski-type form (15) is one of the main structural themes of the manuscript. The first measures the size of the full complex paravector in a positive way, while the second has the mixed sign structure needed for relativistic geometry. Much of the later physics is the disciplined choice of which quadratic form is appropriate in each setting.

Exponentials, projectors, and functional calculus

Exponentials, roots, and functions of simple STA objects follow directly once Eqs. (2) and (10) are fixed.

If $\mathbf{A} \in \mathbb{R}^3$ is written as $\mathbf{A} = \theta \mathbf{u}$ with $\theta = \|\mathbf{A}\|$, $\mathbf{u} \in \mathbb{R}^3$, and $\mathbf{u}^2 = 1$, define the idempotents

$$\Pi(\pm) := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}) \quad (16)$$

satisfying $\Pi(\pm)^2 = \Pi(\pm)$, $\Pi(+)\Pi(-) = 0$, $\Pi(+)+\Pi(-) = 1$, and $(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm\Pi(\pm)$. In projector form, the real and imaginary exponentials are

$$\begin{aligned} \exp(\pm \mathbf{A} \cdot \boldsymbol{\sigma}) &= e^{\pm\theta}\Pi(+)+e^{\mp\theta}\Pi(-) \\ &= \cosh \theta \pm \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma}, \end{aligned} \quad (17)$$

$$\begin{aligned} \exp(\pm i \mathbf{A} \cdot \boldsymbol{\sigma}) &= e^{\pm i\theta}\Pi(+)+e^{\mp i\theta}\Pi(-) \\ &= \cos \theta \pm i \sin \theta \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (18)$$

Equations (17) and (18) underlie the later discussions of functions of $\mathbf{u} \cdot \boldsymbol{\sigma}$, spin decomposition, and the matrix/quaternion bridge. Geometrically, $\Pi(+)$ and $\Pi(-)$ isolate the parts aligned and anti-aligned with the direction $\mathbf{u}$. Later these same idempotents become spin projectors, but their algebraic role is already visible here: they split an object into two sectors adapted to a chosen axis.

For a scalar-valued function $f : \{\pm 1\} \rightarrow \mathbb{C}$,

$$f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(+)\Pi(+)+f(-)\Pi(-). \quad (19)$$

This gives a compact spectral calculus for logarithms, exponentials, and projector-based decompositions. The significance of Eq. (19) is broader than the specific examples used later. Once the minimal polynomial of an STA element is known, functions of that element can be defined without leaving the algebraic framework already established.

Supporting Algebraic Derivations

Eqs. (7), (13), and (19) form the algebraic spine of the manuscript. The derivations collected next record commutators, inverse formulas, logarithms, complex-product decompositions, and functional expansions used later. On a first reading, this subsection may be skimmed and the forward exposition resumed at Section III.

Commutator Algebra

Before specializing to particular STA objects, it is helpful to record the algebraic behavior of the commutator itself, since many later identities are compact ways of saying that certain products commute or fail to commute in controlled ways. See Eqs. (9), (2), and (10). Here $X, Y \in \mathbb{C} \oplus \mathbb{C}^3$ are generic complex paravectors, and $[X, Y] := XY - YX$ is their commutator.

$$\begin{aligned} [X, Y] &:= XY - YX, \\ [Y, X] &= -[X, Y], \\ [X, Y] &= 0 \Rightarrow XY = YX \Rightarrow \text{commutative}. \end{aligned}$$

Vector Notation, Norms, and Cross Products

The elementary Euclidean conventions are recorded here before the sigma basis is brought back in, so that the later STA formulas can always be read against familiar three-vector operations.

$\mathbf{A}$ = 3-element vector,

$\mathbf{B}$ = 3-element vector.

The dot product and squared magnitude are

$$\mathbf{A} \cdot \mathbf{B} = A_1 B_1 + A_2 B_2 + A_3 B_3.$$

$$\mathbf{A}^2 = \mathbf{A} \cdot \mathbf{A}.$$

For a real scalar a , the absolute value is

$$|a| = \sqrt{a^2}.$$

For a complex scalar $S = a + ib$, the usual modulus is

$$\begin{aligned} |S| &= \sqrt{SS^*}, \\ &= \sqrt{(a + ib)(a - ib)}, \\ &= \sqrt{a^2 + b^2}. \end{aligned}$$

The Euclidean norm of a real vector is

$$\|\mathbf{A}\| = \sqrt{\mathbf{A}^2}.$$

For a complex vector $\mathbf{V} = \mathbf{A} + i\mathbf{B}$, the corresponding positive norm is

$$\begin{aligned} \|\mathbf{V}\| &= \sqrt{\mathbf{V}\mathbf{V}^*}, \\ &= \sqrt{(\mathbf{A} + i\mathbf{B}) \cdot (\mathbf{A} - i\mathbf{B})}, \\ &= \sqrt{\mathbf{A}^2 + \mathbf{B}^2}. \end{aligned}$$

Note. The quadratic product $\mathbf{V}^2$ is not the same object as the norm square $\|\mathbf{V}\|^2$.

The cross product is

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= \{A_2B_3 - A_3B_2, -A_1B_3 + A_3B_1, A_1B_2 - A_2B_1\}, \\ \mathbf{B} \times \mathbf{A} &= -\mathbf{A} \times \mathbf{B}. \end{aligned}$$

Sigma Algebra

The sigma generators and their multiplication rules are written out here explicitly so that every later paravector identity can be traced back to the defining algebra. See Eqs. (1) and (2). For $n, m \in \{1, 2, 3\}$,

$$\sigma_n^2 = +1, \sigma_n \sigma_m = -\sigma_m \sigma_n \text{ if } m \neq n.$$

The pseudoscalar is the oriented product of the three generators:

$$\begin{aligned} i &= \sigma_1 \sigma_2 \sigma_3, \\ i^2 &= -1, \\ (\sigma_1 \sigma_2 \sigma_3)^2 &= \sigma_1 \sigma_2 (\sigma_3 \sigma_1) \sigma_2 \sigma_3, \\ &= -\sigma_1 (\sigma_2 \sigma_1) (\sigma_3 \sigma_2) \sigma_3, \\ &= -\sigma_1^2 \sigma_2^2 \sigma_3^2, \\ &= -1. \end{aligned}$$

The bivector sector is

$$\begin{aligned} i\sigma &= \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \\ [i, \sigma_n] &= 0, \\ i\sigma_n &= \sigma_n i. \end{aligned}$$

The Kronecker delta and Levi-Civita symbols are

$$\begin{aligned} \delta_{nm} &= 1 \text{ if } n = m, \\ &= 0 \text{ else.} \end{aligned}$$

$$\begin{aligned} \epsilon_{nmk} &= +1 \text{ if } \{1, 2, 3\}, \{2, 3, 1\}, \{3, 1, 2\}, \\ &\text{even permutation,} \\ &= -1 \text{ if } \{3, 2, 1\}, \{1, 3, 2\}, \{2, 1, 3\}, \\ &\text{odd permutation,} \\ &= 0 \text{ if any two indices are equal.} \end{aligned}$$

With these conventions,

$$\begin{aligned} \sigma_n \sigma_m &= \delta_{nm} + \epsilon_{nmk} i \sigma_k, \\ &= \epsilon_{nmk} i \sigma_k \text{ if } n \neq m. \end{aligned}$$

$$\begin{aligned} \sigma_1 \sigma_2 \sigma_3 &= (\delta_{12} + \epsilon_{123} i \sigma_3) \sigma_3, \\ &= (0 + i \sigma_3) \sigma_3 = i. \end{aligned}$$

The algebra is noncommutative precisely because distinct generators anticommute:

$$\begin{aligned} [\sigma_n, \sigma_m] &= \sigma_n \sigma_m - \sigma_m \sigma_n, \\ &= \sigma_n \sigma_m - (-\sigma_n \sigma_m), \\ &= 2\sigma_n \sigma_m, \\ &= 2\epsilon_{nmk} i \sigma_k. \end{aligned}$$

$$\begin{aligned} [\sigma_2, \sigma_3] &= 2\sigma_1 \sigma_2 \sigma_3 \sigma_1 = 2i\sigma_1, \\ [\sigma_3, \sigma_1] &= 2\sigma_1 \sigma_2 \sigma_3 \sigma_2 = 2i\sigma_2, \\ [\sigma_1, \sigma_2] &= 2\sigma_1 \sigma_2 \sigma_3 \sigma_3 = 2i\sigma_3. \end{aligned}$$

Sigma-Conjugate Action on Sigma Products

The sigma-conjugate identities derived from Eqs. (6)–(8) are expanded here on explicit sigma products.

$$\begin{aligned} \sigma_n^* &= -\sigma_n, \\ i^* &= -i, \\ (i\sigma_n)^* &= i\sigma_n, \\ (\sigma_n \sigma_m)^* &= \sigma_n \sigma_m, \\ (\sigma_1 \sigma_2 \sigma_3)^* &= \sigma_1^* \sigma_2^* \sigma_3^* \quad \text{does not flip order,} \\ &= (-\sigma_1)(-\sigma_2)(-\sigma_3), \\ &= -\sigma_1 \sigma_2 \sigma_3. \end{aligned}$$

$$\begin{aligned} (\sigma_n \sigma_m)^* &= \sigma_n^* \sigma_m^*, \\ &= (-\sigma_n)(-\sigma_m), \\ &= \sigma_n \sigma_m. \end{aligned}$$

$$\begin{aligned} (i\sigma_n)^* &= i^* \sigma_n^*, \\ &= (-i)(-\sigma_n), \\ &= i\sigma_n. \end{aligned}$$

Reverse-Operator Action on Sigma Products

The reverse identities from Eqs. (6)–(8) are expanded here on explicit sigma products.

$$\begin{aligned}
\sigma_n^R &= \sigma_n, \\
i^R &= -i \quad (\sigma_1 \sigma_2 \sigma_3)^R = \sigma_3 \sigma_2 \sigma_1 = -\sigma_1 \sigma_2 \sigma_3, \\
(\sigma_n \sigma_m)^R &= \sigma_m \sigma_n, \\
&= -\sigma_n \sigma_m \text{ if } n \neq m. \\
(i\sigma_n)^R &= -i\sigma_n, \\
(i\sigma_n)^R &= \sigma_n(i)^R, \\
&= \sigma_n(-i), \\
&= -i\sigma_n.
\end{aligned}$$

Grade Bookkeeping

Tables 1 and 2 already summarize the four homogeneous grades and the signs acquired under the sigma-conjugate, reverse, and combined operators. The explicit calculations above supply the derivation behind that bookkeeping, while the tables are the summary used later when paravector products and component formulas are decomposed by grade.

Vector Form and Explicit Sigma Products

Equation (9) is expanded here into the concrete sigma-product rules behind Eq. (10). The purpose is to show explicitly how ordinary vector operations are encoded inside the algebra.

$$\sigma = \{\sigma_1, \sigma_2, \sigma_3\} \quad \text{vector of algebraic objects.}$$

$$\mathbf{A} = 3\text{-element vector,}$$

$$\mathbf{B} = 3\text{-element vector.}$$

The sigma-vector contraction is

$$\mathbf{A} \cdot \sigma = A_1 \sigma_1 + A_2 \sigma_2 + A_3 \sigma_3.$$

Addition remains componentwise:

$$\mathbf{A} \cdot \sigma + \mathbf{B} \cdot \sigma = (\mathbf{A} + \mathbf{B}) \cdot \sigma.$$

Expanding the product gives

$$\begin{aligned}
(\mathbf{A} \cdot \sigma)(\mathbf{B} \cdot \sigma) &= \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \sigma, \\
i &= \sigma_1 \sigma_2 \sigma_3, \\
&= \sum_n \sum_m A_n B_m \sigma_n \sigma_m, \\
&= \sum_n A_n B_n \sigma_n^2, \\
&\quad + \sum_{n < m} (A_n B_m - A_m B_n) \sigma_n \sigma_m, \\
&= \mathbf{A} \cdot \mathbf{B}, \\
&\quad + (A_1 B_2 - A_2 B_1) \sigma_1 \sigma_2,
\end{aligned}$$

$$\begin{aligned}
&\quad + (A_1 B_3 - A_3 B_1) \sigma_1 \sigma_3, \\
&\quad + (A_2 B_3 - A_3 B_2) \sigma_2 \sigma_3, \\
&= \mathbf{A} \cdot \mathbf{B}, \\
&\quad + (A_1 B_2 - A_2 B_1) \sigma_1 \sigma_2 (\sigma_3^2), \\
&\quad + (A_1 B_3 - A_3 B_1) \sigma_1 (\sigma_2^2) \sigma_3, \\
&\quad + (A_2 B_3 - A_3 B_2) (\sigma_1^2) \sigma_2 \sigma_3, \\
&= \mathbf{A} \cdot \mathbf{B}, \\
&\quad + (A_1 B_2 - A_2 B_1) i \sigma_3, \\
&\quad - (A_1 B_3 - A_3 B_1) i \sigma_2, \\
&\quad + (A_2 B_3 - A_3 B_2) i \sigma_1.
\end{aligned}$$

From this one immediately reads off the power and contraction identities

$$\begin{aligned}
(\mathbf{A} \cdot \sigma)^2 &= \mathbf{A}^2, \\
(\mathbf{A} \cdot \sigma)^{2n} &= (\mathbf{A}^2)^n, \\
(\mathbf{A} \cdot \sigma)^{2n+1} &= (\mathbf{A}^2)^n (\mathbf{A} \cdot \sigma), \\
((\mathbf{A} \cdot \sigma)(\mathbf{B} \cdot \sigma))^* &= (\mathbf{A} \cdot \sigma)(\mathbf{B} \cdot \sigma), \\
(\mathbf{A} \cdot \sigma)\sigma &= \mathbf{A} + i\mathbf{A} \times \sigma, \\
\sigma_n \sigma &= (e_n \cdot \sigma)\sigma, \\
&= e_n + i e_n \times \sigma.
\end{aligned}$$

The symmetric and antisymmetric parts recover the Euclidean dot and cross products:

$$\begin{aligned}
(\mathbf{A} \cdot \sigma)(\mathbf{B} \cdot \sigma) &= \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \sigma, \\
(\mathbf{B} \cdot \sigma)(\mathbf{A} \cdot \sigma) &= \mathbf{A} \cdot \mathbf{B} - i(\mathbf{A} \times \mathbf{B}) \cdot \sigma, \\
[\mathbf{A} \cdot \sigma, \mathbf{B} \cdot \sigma] &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \sigma, \\
(\mathbf{A} \times \mathbf{B}) \cdot \sigma &= [\mathbf{A} \cdot \sigma, \mathbf{B} \cdot \sigma]/(2i), \\
\mathbf{A} \cdot \mathbf{B} &= ((\mathbf{A} \cdot \sigma)(\mathbf{B} \cdot \sigma) + (\mathbf{B} \cdot \sigma)(\mathbf{A} \cdot \sigma))/2, \\
[\mathbf{A} \cdot \sigma, \sigma] &= -2i\mathbf{A} \times \sigma = 2i\sigma \times \mathbf{A}, \\
[\mathbf{A} \cdot (i\sigma), \sigma] &= 2\mathbf{A} \times \sigma, \\
[\mathbf{A} \cdot \sigma, \sigma] \cdot \mathbf{B} &= 2i(\sigma \times \mathbf{A}) \cdot \mathbf{B}, \\
&= 2i(\mathbf{A} \times \mathbf{B}) \cdot \sigma, \\
&= [\mathbf{A} \cdot \sigma, \mathbf{B} \cdot \sigma].
\end{aligned}$$

Products of Real Scalars, Vectors, and Paravectors

This subsection expands the scalar, vector, and paravector multiplication rule of Eq. (13).

$$\begin{aligned}
X &= a + \mathbf{A} \cdot \sigma \in \mathbb{R} \oplus \mathbb{R}^3, \\
Y &= b + \mathbf{B} \cdot \sigma \in \mathbb{R} \oplus \mathbb{R}^3, \\
XY &= (a + \mathbf{A} \cdot \sigma)(b + \mathbf{B} \cdot \sigma), \\
&= ab + a(\mathbf{B} \cdot \sigma) + b(\mathbf{A} \cdot \sigma) + (\mathbf{A} \cdot \sigma)(\mathbf{B} \cdot \sigma), \\
&= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + i(\mathbf{A} \times \mathbf{B})) \cdot \sigma, \\
&= YX \text{ if } \mathbf{A} \times \mathbf{B} = 0.
\end{aligned}$$

The product XY generally lies in $\mathbb{R} \oplus \mathbb{C}^3$, and it returns to the real paravector sector precisely when the cross term vanishes:

$$\begin{aligned} XY &\in \mathbb{R} \oplus \mathbb{R}^3 \text{ if } \mathbf{A} \times \mathbf{B} = 0, \\ [X, Y] &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= 0 \text{ if } \mathbf{A} \times \mathbf{B} = 0, \\ &\text{conditionally commutative.} \end{aligned}$$

$$\begin{aligned} (XY)^R &= (Y)^R X^R, \\ (XY)^R &= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} - i(\mathbf{A} \times \mathbf{B})) \cdot \boldsymbol{\sigma}, \\ (XY)^R &= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + i(\mathbf{B} \times \mathbf{A})) \cdot \boldsymbol{\sigma}, \\ &= YX. \end{aligned}$$

$$\begin{aligned} \text{scalar}(XY) &= ab + \mathbf{A} \cdot \mathbf{B}, \\ \text{vector}(XY) &= a\mathbf{B} + b\mathbf{A} + i(\mathbf{A} \times \mathbf{B}), \\ \text{scalar}(YX) &= \text{scalar}(XY), \\ \text{vector}(YX) &= a\mathbf{B} + b\mathbf{A} + i(\mathbf{B} \times \mathbf{A}), \\ &= \text{vector}(XY) \text{ if } \mathbf{A} \times \mathbf{B} = 0. \end{aligned}$$

The square of a real paravector stays in the real sector:

$$\begin{aligned} \text{scalar}(X^2) &= a^2 + \mathbf{A}^2, \\ \text{vector}(X^2) &= 2a\mathbf{A}, \\ X^2 &= a^2 + \mathbf{A}^2 + 2a\mathbf{A} \cdot \boldsymbol{\sigma}, \\ &= b + \mathbf{B} \cdot \boldsymbol{\sigma}, \\ &= Y. \end{aligned}$$

Solving $X^2 = Y$ for X gives

$$\begin{aligned} X &= \pm\sqrt{Y}, \\ a^2 &= \frac{1}{2}(b \pm \sqrt{b^2 - \mathbf{B}^2}), \\ \mathbf{A}^2 &= b - a^2. \end{aligned}$$

The same calculation yields the minimal polynomial and inverse formulas

$$\begin{aligned} X^2 - 2aX + (a^2 - \mathbf{A}^2) &= 0, \\ \langle\langle XY \rangle\rangle^2 &= \langle\langle Y \rangle\rangle^2 \langle\langle X \rangle\rangle^2 \quad \text{if } \mathbf{A} \times \mathbf{B} = 0, \\ X^{-1} &= X^*/(XX^*), \\ &= (a - \mathbf{A} \cdot \boldsymbol{\sigma})/(a^2 - \mathbf{A}^2), \\ &\text{if } a^2 \neq \mathbf{A}^2. \end{aligned}$$

$$\begin{aligned} X^{-1}X &= 1, \\ d(X^{-1}) &= -X^{-1}(dX)X^{-1}, \\ X^{-1}Y &= YX^{-1} \text{ if } \mathbf{A} \times \mathbf{B} = 0. \end{aligned}$$

Exponential of a Real Vector

The exponential becomes especially transparent once the unit-direction element $\mathbf{u} \cdot \boldsymbol{\sigma}$ is resolved into its two eigenprojectors. Write $\mathbf{A} = \theta\mathbf{u}$ with $\theta = \|\mathbf{A}\|$ and $\mathbf{u}^2 = 1$. Then the projectors from Eq. (16) satisfy

$$\begin{aligned} \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \mathbf{u} \cdot \boldsymbol{\sigma} &= \Pi(+)-\Pi(-), \\ 1 &= \Pi(+)+\Pi(-), \\ (\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) &= \pm\Pi(\pm). \end{aligned}$$

Applying the spectral decomposition to the exponential gives the clean projector form

$$\begin{aligned} \exp(\pm\mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\pm\theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= e^{\pm\theta}\Pi(+)+e^{\mp\theta}\Pi(-) \\ &= \cosh(\theta) \pm \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}, \end{aligned}$$

with

$$\begin{aligned} \gamma &= \cosh(\theta), \\ \mathbf{v} &= \tanh(\theta) \mathbf{u}. \end{aligned}$$

In particular,

$$\begin{aligned} \exp(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\ \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) &= \gamma(1 - \mathbf{v} \cdot \boldsymbol{\sigma}). \end{aligned}$$

For a real paravector $X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$, the scalar part commutes with the vector part, so

$$\begin{aligned} \exp(X) &= \exp(a) \exp(\mathbf{A} \cdot \boldsymbol{\sigma}), \\ \exp(X^*) &= \exp(a) \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) = \exp(X)^*, \\ \exp(-X) &= \exp(-a) \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) = \exp(X)^{-1}. \end{aligned}$$

The same projector split also gives the compact trigonometric form of the imaginary exponential:

$$\begin{aligned} \exp(\pm i\mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\pm i\theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= e^{\pm i\theta}\Pi(+)+e^{\mp i\theta}\Pi(-) \\ &= \cos(\theta) \pm i \sin(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned}$$

For the minus sign this is equivalently

$$\exp(-i\mathbf{A} \cdot \boldsymbol{\sigma}) = \cos(\theta) + \sin(\theta) (\mathbf{u} \cdot \mathbf{q}) \quad \text{see quaternions.}$$

Finally, the sigma-conjugate simply flips the projector weights:

$$\begin{aligned} \exp(\mathbf{A} \cdot \boldsymbol{\sigma})^* &= e^{-\theta}\Pi(+)+e^{\theta}\Pi(-) \\ &= \gamma(1 - \mathbf{v} \cdot \boldsymbol{\sigma}) \\ &= \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= \exp((\mathbf{A} \cdot \boldsymbol{\sigma})^*) \\ &= \exp(\mathbf{A} \cdot \boldsymbol{\sigma})^{-1}. \end{aligned}$$

Logarithm of a Real Paravector

The logarithm is the inverse problem for the same spectral calculus. Let

$$Y = b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3,$$

and assume $b > \|\mathbf{B}\|$, so that the two real eigenvalues are positive and the principal logarithm is defined. Writing $\mathbf{B} = \|\mathbf{B}\|\mathbf{u}$ when $\mathbf{B} \neq 0$, one has

$$\begin{aligned}\lambda(\pm) &= b \pm \|\mathbf{B}\|, \\ Y &= \lambda(+)\Pi(+) + \lambda(-)\Pi(-),\end{aligned}$$

with the same projector pair $\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$. The logarithm therefore acts separately on the two eigenvalues:

$$\begin{aligned}\log(Y) &= \log(\lambda(+))\Pi(+) + \log(\lambda(-))\Pi(-) \\ &= \frac{1}{2} \log(\lambda(+)\lambda(-)) + \frac{1}{2} \log(\lambda(+)/\lambda(-)) \mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned}$$

If $Y = \exp(X)$ with $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ and $\mathbf{A} = \alpha\mathbf{u}$, then $\lambda(\pm) = e^{a \pm \alpha}$, so the recovered parameters are

$$\begin{aligned}a &= \frac{1}{2} \log(b^2 - \mathbf{B}^2), \\ \alpha &= \operatorname{arctanh}(\|\mathbf{B}\|/b), \\ \mathbf{A} &= \alpha \mathbf{u} = \operatorname{arctanh}(\|\mathbf{B}\|/b) \frac{\mathbf{B}}{\|\mathbf{B}\|}.\end{aligned}$$

Thus the logarithm of a time-like real paravector is

$$\log(Y) = a + \mathbf{A} \cdot \boldsymbol{\sigma}.$$

Complex Paravectors

Writing the generic complex paravector as in Eq. (11), one has

$$\begin{aligned}i &= \sigma_1 \sigma_2 \sigma_3, \\ X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\ &= X + iY, \\ &= (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} \quad \text{paravector.} \\ a &= \operatorname{scalar}(X) \in \mathbb{R}, \\ \mathbf{A} &= \operatorname{vector}(X) \in \mathbb{R}^3, \\ b &= \operatorname{scalar}(Y) \in \mathbb{R}, \\ \mathbf{B} &= \operatorname{vector}(Y) \in \mathbb{R}^3, \\ S &= \operatorname{scalar}(Z) \in \mathbb{C} \quad \text{complex scalar,} \\ \mathbf{V} &= \operatorname{vector}(Z) \in \mathbb{C}^3 \quad \text{complex vector,} \\ Z^* &= (a - bi) + (-\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ Z^R &= (a - bi) + (\mathbf{A} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ Z^{R*} &= (a + bi) - (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ X &= \operatorname{real}(Z) = (Z + Z^R)/2, \\ Y &= \operatorname{img}(Z) = (Z - Z^R)/(2i), \\ S &= a + bi, \\ &= (Z + Z^{R*})/2.\end{aligned}$$

$$\begin{aligned}\mathbf{V} \cdot \boldsymbol{\sigma} &= (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= (Z - Z^{R*})/2.\end{aligned}$$

Products of Complex Paravectors

$$\begin{aligned}X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\ &= X + iY.\end{aligned}$$

$$\operatorname{scalar}(ZZ') = \operatorname{scalar}(Z'Z) \quad \text{cyclic property.}$$

In particular, if $Z' = MZ^{*R}$, then

$$\begin{aligned}\operatorname{scalar}(ZMZ^{*R}) &= \operatorname{scalar}(MZ^{*R}Z), \\ &= \operatorname{scalar}(M)(Z^{*R}Z).\end{aligned}$$

Here $Z^{*R}Z$ is scalar.

$$\begin{aligned}\operatorname{scalar}(ZZ^*) &= SS^* - \mathbf{V}\mathbf{V}^* \quad \text{real,} \\ &= (a^2 + b^2) - (\mathbf{A}^2 + \mathbf{B}^2).\end{aligned}$$

$$\begin{aligned}\|Z\|^2 &= |S|^2 + \|\mathbf{V}\|^2, \\ &= SS^* + \mathbf{V}\mathbf{V}^*, \\ &= (a^2 + b^2) + (\mathbf{A}^2 + \mathbf{B}^2) \quad \text{non-negative real,} \\ &= \operatorname{scalar}(Z^R Z), \\ &= \operatorname{scalar}(ZZ^R), \\ &= \frac{1}{2}(Z^R Z + (Z^R Z)^{*R}), \\ &= \frac{1}{2}(Z^R Z + Z^{*R} Z^*).\end{aligned}$$

$$\begin{aligned}\|\mathbf{r} \exp(i\theta)Z\|^2 &= \mathbf{r}^2 \|Z\|^2, \\ ZZ^* &= \text{real scalar} + \text{complex vector}, \\ ZZ^R &= \text{real scalar} + \text{real vector}.\end{aligned}$$

The Minkowski-type quadratic form ZZ^{R*} is always a complex scalar, so the involution relations reduce to

$$\begin{aligned}Z^* Z^R &= Z^R Z^* = (Z^{R*} Z)^*, \\ Z^* Z &= ZZ^* \text{ if } \mathbf{A} \times \mathbf{B} = 0, \\ Z^R Z &= ZZ^R \text{ if } \mathbf{A} \times \mathbf{B} = 0, \\ (ZZ')^* &= Z^*(Z')^*, \\ (ZZ')^R &= (Z')^R Z^R, \\ (ZZ')^{*R} &= (Z')^{*R} Z^{*R}.\end{aligned}$$

For differential expressions, the product rule takes the form

$$\begin{aligned}\partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\ \partial(ZZ') &= (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}} Z' .\end{aligned}$$

Paravector Metric and Interval Form

The two quadratic forms introduced in Eqs. (14) and (15) take the following detailed form: For a real scalar s and real vector $\mathbf{v}$,

$$\begin{aligned}\langle\langle s + \mathbf{v} \cdot \boldsymbol{\sigma} \rangle\rangle^2 &:= s^2 - \mathbf{v}^2, \\ &= (s + \mathbf{v} \cdot \boldsymbol{\sigma})(s - \mathbf{v} \cdot \boldsymbol{\sigma}), \\ &= (s - \mathbf{v} \cdot \boldsymbol{\sigma})(s + \mathbf{v} \cdot \boldsymbol{\sigma}).\end{aligned}$$

$$\begin{aligned}X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\ &= X + iY.\end{aligned}$$

$$\begin{aligned}\langle\langle Z \rangle\rangle^2 &= S^2 - \mathbf{V}^2 \quad \text{complex scalar}, \\ &= ZZ^{\mathbf{R}*} = Z^{\mathbf{R}*}Z, \\ &= (a + ib)^2 - (\mathbf{A} + i\mathbf{B})^2, \\ &= (a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2) + 2i(ab - \mathbf{A} \cdot \mathbf{B}).\end{aligned}$$

$$\begin{aligned}\langle\langle \mathbf{r} \exp(i\theta) Z \rangle\rangle^2 &= \mathbf{r}^2 \exp(2i\theta) \langle\langle Z \rangle\rangle^2, \\ \langle\langle X \rangle\rangle^2 &= a^2 - \mathbf{A}^2 \quad \text{real scalar}, \\ &= XX^* = X^*X.\end{aligned}$$

Inverse Formulas

For $Z \in \mathbb{C} \oplus \mathbb{C}^3$, the inverse is controlled by the Minkowski-type quadratic form:

$$\begin{aligned}Z^{-1} &= Z^{\mathbf{R}*} / \langle\langle Z \rangle\rangle^2, \\ &= Z^{\mathbf{R}*} / (Z^{\mathbf{R}*}Z).\end{aligned}$$

If $Z^{\mathbf{R}*}Z = 0$, then the inverse does not exist.

Minimal Polynomial and Spectral Decomposition

The spectral data of a complex paravector are collected here because the later functional calculus, projector formulas, and exponential identities all depend on this algebraic factorization. See Eq. (19).

$$\begin{aligned}Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\ Z^2 - 2SZ + (S^2 - \mathbf{V}^2) &= 0, \\ \lambda(\pm) &= S \pm \sqrt{\mathbf{V}^2}, \\ (Z - \lambda(-))(Z - \lambda(+)) &= 0.\end{aligned}$$

Exponentials of Complex Paravectors

$$\begin{aligned}X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3, \\ Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{C} \oplus \mathbb{C}^3, \\ &= X + iY.\end{aligned}$$

$$\begin{aligned}\exp(Z) &= \exp(S + \mathbf{V} \cdot \boldsymbol{\sigma}), \\ &= \exp(S) \exp(\mathbf{V} \cdot \boldsymbol{\sigma}).\end{aligned}$$

$$\begin{aligned}\exp(Z^*) &= \exp(S^*) \exp(-\mathbf{V}^* \cdot \boldsymbol{\sigma}), \\ \exp(Z)^{-1} &= \exp(-S) \exp(-\mathbf{V} \cdot \boldsymbol{\sigma}), \\ \exp(Z^*)^{-1} &= \exp(-S^*) \exp(\mathbf{V}^* \cdot \boldsymbol{\sigma}), \\ (\mathbf{V} \cdot \boldsymbol{\sigma})^2 &= \mathbf{V} \cdot \mathbf{V} = \mathbf{V}^2, \\ &= \mathbf{A}^2 - \mathbf{B}^2 + 2i\mathbf{A} \cdot \mathbf{B}.\end{aligned}$$

$$(\mathbf{V} \cdot \boldsymbol{\sigma})^{2n} = (\mathbf{V}^2)^n,$$

$$z = \sqrt{\mathbf{V}^2} \quad \text{complex scalar},$$

$$\begin{aligned}\exp(\mathbf{V} \cdot \boldsymbol{\sigma}) &= \sum_n (\mathbf{V} \cdot \boldsymbol{\sigma})^n / n!, \\ &= \sum_n (\mathbf{V}^2)^n / (2n)! + \sum_n (\mathbf{V}^2)^n / (2n+1)! (\mathbf{V} \cdot \boldsymbol{\sigma}), \\ &= \cosh(z) + \sinh(z) (\mathbf{V} \cdot \boldsymbol{\sigma}) / z.\end{aligned}$$

Complex Field Decomposition

$$\partial = \partial_t + \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma},$$

$$F = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma},$$

$$\mathbf{a} : \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{R},$$

$$\mathbf{A} : \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{R}^3,$$

$$\mathbf{b} : \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{R},$$

$$\mathbf{B} : \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{R}^3,$$

$$F : \mathbb{R} \times \mathbb{R}^3 \Rightarrow \mathbb{C} \oplus \mathbb{C}^3.$$

Functions of a Unit-Direction Element

The projector pair $\Pi(\pm)$ is the spectral decomposition of the unit-direction element $\mathbf{u} \cdot \boldsymbol{\sigma}$. The corresponding eigenvalue data are

$$\begin{aligned}\Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \Pi(\pm)^2 &= \Pi(\pm), \\ \Pi(+)\Pi(-) &= 0, \\ \Pi(+) + \Pi(-) &= 1, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) &= \pm\Pi(\pm),\end{aligned}$$

so $\mathbf{u} \cdot \boldsymbol{\sigma}$ has the two eigenvalues $+1$ and -1 . Any function with a convergent power series therefore collapses to those two scalar values:

$$\begin{aligned}f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(+1)\Pi(+) + f(-1)\Pi(-) \\ &= \frac{1}{2}(f(+1) + f(-1)) + \frac{1}{2}(f(+1) - f(-1)) \mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned}$$

This is the basic projector calculus for a unit-direction element. Its common special cases are

$$\begin{aligned}\sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} &= \pm \Pi(+) \pm i \Pi(-), \\ \log(\mathbf{u} \cdot \boldsymbol{\sigma}) &= 2\pi i \left(n \Pi(+) + \left(m + \frac{1}{2}\right) \Pi(-) \right), \\ (z + \mathbf{u} \cdot \boldsymbol{\sigma})^{-1} &= \frac{1}{z+1} \Pi(+) + \frac{1}{z-1} \Pi(-) \\ &= \frac{z - \mathbf{u} \cdot \boldsymbol{\sigma}}{z^2 - 1}, \\ \frac{i - \mathbf{u} \cdot \boldsymbol{\sigma}}{i + \mathbf{u} \cdot \boldsymbol{\sigma}} &= i \mathbf{u} \cdot \boldsymbol{\sigma}, \\ \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) &= e^\theta \Pi(+) + e^{-\theta} \Pi(-), \\ \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}) &= e^{i\theta} \Pi(+) + e^{-i\theta} \Pi(-),\end{aligned}$$

where $m, n \in \mathbb{Z}$ select the branch of the logarithm.

Functions of a Paravector

Let

$$\begin{aligned}X &= t + \mathbf{r} \cdot \boldsymbol{\sigma}, \\ t &= \text{scalar}(X) \in \mathbb{R}, \\ \mathbf{r} &= \text{vector}(X) \in \mathbb{R}^3.\end{aligned}$$

If $\mathbf{r} \neq 0$, write $\mathbf{r} = \|\mathbf{r}\| \mathbf{u}$ with $\mathbf{u}^2 = 1$. Then the minimal polynomial of X is

$$\begin{aligned}X^2 - 2tX + (t^2 - \mathbf{r}^2) &= 0, \\ \lambda(\pm) &= t \pm \|\mathbf{r}\|, \\ 0 &= (X - \lambda(-))(X - \lambda(+)),\end{aligned}$$

so X has the projector decomposition

$$\begin{aligned}\Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ X &= \lambda(+) \Pi(+) + \lambda(-) \Pi(-), \\ \Pi(\pm) X &= \lambda(\pm) \Pi(\pm).\end{aligned}$$

Any analytic function of X is therefore obtained by applying the scalar function to the two eigenvalues:

$$\begin{aligned}f(X) &= f(\lambda(+)) \Pi(+) + f(\lambda(-)) \Pi(-) \\ &= \frac{1}{2} \left(f(\lambda(+)) + f(\lambda(-)) \right) \\ &\quad + \frac{1}{2} \left(f(\lambda(+)) - f(\lambda(-)) \right) \mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned}$$

If $f(X) = t' + \mathbf{r}' \cdot \boldsymbol{\sigma}$, then

$$\begin{aligned}t' &= \frac{1}{2} \left(f(\lambda(+)) + f(\lambda(-)) \right), \\ \|\mathbf{r}'\| &= \frac{1}{2} \left(f(\lambda(+)) - f(\lambda(-)) \right), \\ \mathbf{r}' &= \|\mathbf{r}'\| \mathbf{u}.\end{aligned}$$

The main special cases are

$$\begin{aligned}\exp(X) &= e^{\lambda(+)} \Pi(+) + e^{\lambda(-)} \Pi(-) \\ &= e^t (\cosh(\|\mathbf{r}\|) + \sinh(\|\mathbf{r}\|) \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \sqrt{X} &= \pm \sqrt{\lambda(+)} \Pi(+) \pm \sqrt{\lambda(-)} \Pi(-), \\ \log(X) &= \log(\lambda(+)) \Pi(+) + \log(\lambda(-)) \Pi(-), \\ X^{-1} &= \frac{\Pi(+)}{\lambda(+)} + \frac{\Pi(-)}{\lambda(-)} \\ &= \frac{X^*}{t^2 - \mathbf{r}^2}, \\ X^n &= \lambda(+)^n \Pi(+) + \lambda(-)^n \Pi(-).\end{aligned}$$

When two different paravectors $X = t + \|\mathbf{r}\| \mathbf{u} \cdot \boldsymbol{\sigma}$ and $X' = t' + \|\mathbf{r}'\| \mathbf{u}' \cdot \boldsymbol{\sigma}$ are decomposed along different axes, their functions multiply by combining the two projector families:

$$f(X)g(X') = \sum_{s, s' = \pm} f(\lambda(s)) g(\lambda'(s')) \Pi(s) \Pi'(s').$$

The mixed projector products are

$$\Pi(s) \Pi'(s') = \frac{1}{4} \left(1 + ss' \mathbf{u} \cdot \mathbf{u}' + (s\mathbf{u} + s'\mathbf{u}' + ss' i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma} \right),$$

with $s, s' \in \{+1, -1\}$. This compact formula reproduces all four explicit same-side and cross-side products. As a useful example, the Cayley-type transform is itself a projector exponential:

$$\begin{aligned}\frac{1 - \mathbf{r} \cdot \boldsymbol{\sigma}}{1 + \mathbf{r} \cdot \boldsymbol{\sigma}} &= \exp(-2 \operatorname{arctanh}(\mathbf{r} \cdot \boldsymbol{\sigma})) \\ &= e^{-2 \operatorname{arctanh}(\|\mathbf{r}\|)} \Pi(+) + e^{+2 \operatorname{arctanh}(\|\mathbf{r}\|)} \Pi(-).\end{aligned}$$

Supporting Tables

The accompanying tables collect the remaining recovery formulas, product decompositions, and component expansions used throughout the algebraic development. The remaining algebra tables collect the recovery formulas, product decompositions, and order-dependent component expansions used in later calculations.

Table 3. Components of Z .

scalar:	$a + ib = (Z + Z^{\mathbf{R}*})/2$
vector:	$(\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} = (Z - Z^{\mathbf{R}*})/2$
real part:	$a + \mathbf{A} \cdot \boldsymbol{\sigma} = (Z + Z^{\mathbf{R}})/2$
imaginary part:	$b + \mathbf{B} \cdot \boldsymbol{\sigma} = (Z - Z^{\mathbf{R}})/(2i)$
real scalar:	$a = (Z + Z^* + Z^{\mathbf{R}} + Z^{\mathbf{R}*})/4$
imaginary scalar:	$b = (Z - Z^* - Z^{\mathbf{R}} + Z^{\mathbf{R}*})/(4i)$
real vector:	$\mathbf{A} \cdot \boldsymbol{\sigma} = (Z - Z^* + Z^{\mathbf{R}} - Z^{\mathbf{R}*})/4$
imaginary vector:	$\mathbf{B} \cdot \boldsymbol{\sigma} = (Z + Z^* - Z^{\mathbf{R}} - Z^{\mathbf{R}*})/(4i)$

Tables 4 and 5 collect the full component expansions for the generic product ZZ' and the special case Z^2 .

Table 4. Components of ZZ' .

real scalar:	$aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}'$
imaginary scalar:	$ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}'$
real vector:	$a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}'$
imaginary vector:	$a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'$

Table 5. Components of Z^2 .

real scalar:	$a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2$	
imaginary scalar:	$2ab + 2\mathbf{A} \cdot \mathbf{B}$	
real vector:	$2(a\mathbf{A} - b\mathbf{B})$	
imaginary vector:	$2(a\mathbf{B} + b\mathbf{A})$	

Tables 6 and 7 make the order dependence of ZZ^* and Z^*Z explicit.

Table 6. Components of ZZ^* .

real scalar:	$a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2$	
imaginary scalar:	0	
real vector:	$-2\mathbf{A} \times \mathbf{B}$	$\leftarrow$ differs from Z^*Z
imaginary vector:	$2(a\mathbf{B} - b\mathbf{A})$	

Table 7. Components of Z^*Z .

real scalar:	$a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2$	
imaginary scalar:	0	
real vector:	$2\mathbf{A} \times \mathbf{B}$	$\leftarrow$ differs from ZZ^*
imaginary vector:	$2(a\mathbf{B} - b\mathbf{A})$	

Tables 8 and 9 play the same role for products involving the reverse operator.

Table 8. Components of ZZ^R .

real scalar:	$a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2$	
imaginary scalar:	0	
real vector:	$2(a\mathbf{A} + b\mathbf{B} + \mathbf{A} \times \mathbf{B})$	$\leftarrow$ differs from $Z^R Z$
imaginary vector:	$\mathbf{0}$	

Table 9. Components of $Z^R Z$.

real scalar:	$a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2$	
imaginary scalar:	0	
real vector:	$2(a\mathbf{A} + b\mathbf{B} - \mathbf{A} \times \mathbf{B})$	$\leftarrow$ differs from ZZ^R
imaginary vector:	$\mathbf{0}$	

Tables 10 and 11 show that the conjugate-reverse products collapse to purely scalar and imaginary-scalar parts.

Table 10. Components of ZZ^R .

real scalar:	$a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2$	
imaginary scalar:	$2ab - 2\mathbf{A} \cdot \mathbf{B}$	
real vector:	$\mathbf{0}$	
imaginary vector:	$\mathbf{0}$	

Table 11. Components of $Z^R Z$.

real scalar:	$a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2$	
imaginary scalar:	$2ab - 2\mathbf{A} \cdot \mathbf{B}$	
real vector:	$\mathbf{0}$	
imaginary vector:	$\mathbf{0}$	

III. SPACE-TIME KINEMATICS

With the algebraic foundations in place, the same paravector structure can now be read as space-time. The transition from algebra to geometry is immediate: the quadratic form becomes the interval, and normalized time-like paravectors become proper velocities.

Differential space-time and proper velocity

Specializing the real paravector $X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$ from Eq. (12) to differentials gives

$$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma} \quad (20)$$

and the interval is the differential version of Eq. (15):

$$ds^2 = \langle\langle dX \rangle\rangle^2 = dX dX^* = dt^2 - d\mathbf{r}^2. \quad (21)$$

Here $t, s, \gamma \in \mathbb{R}$, $\mathbf{r}, \mathbf{v}, \mathbf{P} \in \mathbb{R}^3$, and $X, U \in \mathbb{R} \oplus \mathbb{R}^3$. The scalar part dt is the temporal separation and the vector part $d\mathbf{r} \cdot \boldsymbol{\sigma}$ is the spatial separation. Equation (21) is therefore not an added metric axiom but the direct spacetime interpretation of the quadratic form already introduced in Eq. (15).

If $\mathbf{v} = d\mathbf{r}/dt \in \mathbb{R}^3$ and $\gamma = (1 - \mathbf{v}^2)^{-1/2} \in \mathbb{R}$, then

$$U := \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad \langle\langle U \rangle\rangle^2 = 1. \quad (22)$$

Multiplying Eq. (22) by the real rest mass $m \in \mathbb{R}$ gives the proper momentum formula

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}, \quad E = m\gamma, \quad \mathbf{P} = m\gamma\mathbf{v}, \quad (23)$$

which immediately yields $E^2 - \mathbf{P}^2 = m^2$. The normalization $\langle\langle U \rangle\rangle^2 = 1$ is the essential point: proper velocity is the unit time-like paravector tangent to the worldline. Multiplying by m then produces the energy-momentum paravector in a way that makes the relativistic mass shell relation appear immediately.

Proper acceleration and free motion

The proper acceleration is the real paravector

$$A := \frac{dU}{ds} = \frac{d^2 X}{ds^2}, \quad \text{scalar}(U^* A) = 0. \quad (24)$$

The orthogonality condition in Eq. (24) is the paravector analogue of the standard four-acceleration orthogonality relation. It is included here because the Lorentz-force law later in the paper should be read as a specific dynamical realization of this general kinematic structure.

A free-particle Lagrangian is

$$\mathcal{L}_{\text{free}} = -\frac{m}{\gamma}, \quad (25)$$

whose Euler-Lagrange equations reproduce inertial motion and the usual nonrelativistic expansion

$$\mathcal{L}_{\text{free}} = -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4). \quad (26)$$

Equation (26) is worth displaying explicitly because it shows how the relativistic action reduces to the classical kinetic expression plus an additive rest-energy constant when $\|\mathbf{v}\| \ll 1$.

Supporting Kinematic Derivations

The derivations collected next expand Eqs. (21), (22), and (23). They keep the later dynamical equations tied to previously defined spacetime objects. On a first reading, this subsection may be skimmed and the forward exposition resumed at Section IV.

Spacetime Paravectors and Interval Recovery

Starting from the displacement and interval formulas in Eqs. (20) and (21), the real paravector

$$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, \quad dX^* = dt - d\mathbf{r} \cdot \boldsymbol{\sigma},$$

is the algebraic form of an infinitesimal space-time displacement. The temporal and spatial pieces are recovered directly by

$$\frac{dX + dX^*}{2} = dt, \quad \frac{dX - dX^*}{2} = d\mathbf{r} \cdot \boldsymbol{\sigma}.$$

Multiplying the displacement by its conjugate gives the interval again,

$$\langle\langle dX \rangle\rangle^2 = dX dX^* = dt^2 - d\mathbf{r}^2 = ds^2.$$

This is the precise point at which the algebraic Minkowski form from Eq. (15) becomes the space-time metric of Eq. (21). For a non-null displacement, the inverse exists and is

$$(dX)^{-1} = \frac{dX^*}{ds^2}.$$

Null displacements are singled out algebraically because this inverse breaks down exactly when $ds^2 = 0$.

Interval Classes and Proper Time

Equation (21) gives the three standard interval classes without any additional geometric machinery:

$$\begin{aligned} ds^2 > 0 &\Rightarrow \text{time-like,} \\ ds^2 = 0 &\Rightarrow \text{null,} \\ ds^2 < 0 &\Rightarrow \text{space-like.} \end{aligned}$$

For a time-like worldline, the parameter s is the proper time in natural units. Writing $\mathbf{v} = d\mathbf{r}/dt$ then gives

$$ds = \sqrt{1 - \mathbf{v}^2} dt, \quad \tau = \int ds = \int \sqrt{1 - \mathbf{v}^2} dt.$$

In SI units the same relation is $ds = c d\tau$. The advantage of the paravector notation is that the interval classification and the proper-time formula both come from the single algebraic identity $dX dX^* = dt^2 - d\mathbf{r}^2$.

Proper Velocity and Proper Acceleration in Components

Starting from Eq. (22), the proper-velocity paravector can be read componentwise as

$$\begin{aligned} U &= \frac{dX}{ds} = \gamma + \gamma \mathbf{v} \cdot \boldsymbol{\sigma}, \\ \text{scalar}(U) &= \gamma, \quad \text{vector}(U) = \gamma \mathbf{v}. \end{aligned}$$

Its normalization $\langle\langle U \rangle\rangle^2 = 1$ implies $U^{-1} = U^*$. Differentiating with respect to s yields

$$\text{scalar}(U^* A) = 0, \quad A := \frac{dU}{ds},$$

so proper acceleration is Minkowski-orthogonal to proper velocity.

If $\mathbf{a} = d\mathbf{v}/dt$, then

$$A = \gamma^4(\mathbf{v} \cdot \mathbf{a}) + \left(\gamma^2 \mathbf{a} + \gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} \right) \cdot \boldsymbol{\sigma}.$$

Its quadratic form is

$$\langle\langle A \rangle\rangle^2 = -\gamma^4 \left(\mathbf{a}^2 + \gamma^2(\mathbf{v} \cdot \mathbf{a})^2 \right),$$

which is strictly space-like unless the acceleration vanishes. Splitting $\mathbf{a}$ into parts parallel and perpendicular to $\mathbf{v}$ makes the same structure more explicit:

$$\langle\langle A \rangle\rangle^2 = -\gamma^4 \left(\mathbf{a}_\perp^2 + \gamma^2 \mathbf{a}_\parallel^2 \right).$$

This is the geometric content later used by the Lorentz-force law.

Proper Momentum and the Mass Shell

Multiplying proper velocity by the rest mass produces the energy-momentum paravector,

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}, \quad E = m\gamma, \quad \mathbf{P} = m\gamma \mathbf{v}.$$

At rest one has $\mathbf{v} = 0$, hence $E = m$ in natural units and $E = mc^2$ in SI units. The kinetic energy is therefore $E - m$.

Because $\langle\langle U \rangle\rangle^2 = 1$, the mass shell follows immediately:

$$\langle\langle mU \rangle\rangle^2 = (E + \mathbf{P} \cdot \boldsymbol{\sigma})(E - \mathbf{P} \cdot \boldsymbol{\sigma}) = E^2 - \mathbf{P}^2 = m^2.$$

For $m = 0$, this reduces to $E^2 = \mathbf{P}^2$, which is the null case appropriate to massless motion.

Action Principle, Hamiltonian, and Low-Velocity Expansion

The free action is

$$S_{\text{free}} = -m \int ds = \int \mathcal{L}_{\text{free}} dt, \quad \mathcal{L}_{\text{free}} = -\frac{m}{\gamma}.$$

Differentiating with respect to $\mathbf{v}$ gives the canonical momentum again,

$$\frac{\partial \mathcal{L}_{\text{free}}}{\partial \mathbf{v}} = m\gamma \mathbf{v} = \mathbf{P}.$$

The Hamiltonian is then

$$\mathcal{H} = \mathbf{v} \cdot \frac{\partial \mathcal{L}_{\text{free}}}{\partial \mathbf{v}} - \mathcal{L}_{\text{free}} = m\gamma \mathbf{v}^2 + \frac{m}{\gamma} = m\gamma = E,$$

so the action principle reproduces both the relativistic momentum and the relativistic energy inside one calculation.

The low-velocity series

$$\begin{aligned} \mathcal{L}_{\text{free}} &= -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4), \\ \mathbf{P} &= m\mathbf{v} + O(\mathbf{v}^3), \\ \mathcal{H} &= m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4) \end{aligned}$$

shows exactly how the relativistic action contains the Newtonian kinetic term together with the additive rest-energy constant. Classical mechanics therefore appears as the approximation, not as the starting point.

Supporting Tables

The accompanying tables record the scalar and vector parts of the proper-velocity and proper-acceleration formulas used in the kinematic discussion. Table 12 resolves the proper-velocity paravector into its scalar and spatial components.

Table 12. Components of dX/ds .

scalar:	γ
vector:	$\gamma \mathbf{v}$

Tables 13 and 14 compare the products UA and U^*A , making the orthogonality structure of proper acceleration explicit.

Table 13. Components of UA .

real scalar:	$2\gamma^5(\mathbf{v} \cdot \mathbf{a})$
imaginary scalar:	0
real vector:	$\gamma^3 \mathbf{a} + 2\gamma^5(\mathbf{v} \cdot \mathbf{a})\mathbf{v}$
imaginary vector:	$\gamma^3(\mathbf{v} \times \mathbf{a})$

Table 14. Components of U^*A .

real scalar:	0
imaginary scalar:	0
real vector:	$\gamma^3 \mathbf{a}$
imaginary vector:	$-\gamma^3(\mathbf{v} \times \mathbf{a})$

IV. ROTORS, BOOSTS, AND COVARIANT TRANSFORMATIONS

Exponentials appear not only as algebraic functions but also as generators of transformations. After interval structure is established, rotors and boosts are the cleanest next step because they explain how the same objects move between frames before fields and wave equations are introduced.

General transform law, spatial rotors, and boosts

A guiding principle of STA is that transformations should be expressed by the same algebra that describes the objects being transformed. Instead of introducing a separate matrix formalism at this stage, one uses exponentials inside the algebra itself to generate rotations and boosts.

A general transform law is

$$X' = T^{-1}XT. \quad (27)$$

Here T denotes a generic invertible transformation element. For a real angle or rapidity $\theta \in \mathbb{R}$ and a unit real vector $\mathbf{u} \in \mathbb{R}^3$, the specific choices used throughout the remainder of the manuscript are R for spatial rotations and L for Lorentz boosts. Two important special cases are the spatial rotor

$$R(\theta, \mathbf{u}) = \exp\left(\frac{\theta}{2} \mathbf{u} \cdot i\boldsymbol{\sigma}\right), \quad X' = R^R X R, \quad (28)$$

and the Lorentz boost

$$L(\theta, \mathbf{u}) = \exp\left(\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right), \quad X' = L^* X L. \quad (29)$$

The symbols R and L are invertible algebra elements rather than vectors or paravectors, while T is reserved for the generic transform law (27). Equation (28) produces Rodrigues' rotation formula, while Eq. (29) produces the usual split into components parallel and perpendicular to the boost axis [2, 3, 9]. The formal similarity between Eqs. (28) and (29) is deliberate. Both are exponential transformations, but bivector-like generators produce trigonometric rotation structure while vector-like generators produce hyperbolic boost structure. The contrast shows how Euclidean and Lorentzian motions coexist inside one algebraic language.

Supporting Transformation Derivations

The formulas collected here expand Eqs. (27)–(29) into explicit basis transforms, boost parameterizations, and composition rules. On a first reading, this subsection may be skimmed and the forward exposition resumed at Section V.

General Transformation Identities

Equation (27) is expanded here into differential and commutator form.

$$\begin{aligned} X' &= T^{-1}XT, \\ &= T^{-1}(t + \mathbf{r} \cdot \boldsymbol{\sigma})T, \\ &= t + \mathbf{r} \cdot (T^{-1}\boldsymbol{\sigma}T), \\ &= t + \mathbf{r} \cdot \boldsymbol{\sigma}'. \end{aligned}$$

$$T^{-1} = T^{R*}/(T^{R*}T).$$

For boosts and rotors one has $T^{R*}T = 1$, so the inverse simplifies to

$$\begin{aligned} T^{R*}T &= 1, \quad T^{-1} = T^{R*}, \\ T^{-1}T &= 1, \\ d(T^{-1})T &= -T^{-1}dT, \\ dX' &= d(T^{-1})XT + T^{-1}dXT + T^{-1}XdT, \\ &= T^{-1}dXT. \end{aligned}$$

Inserting TT^{-1} into the differentiated expression isolates the commutator contribution:

$$\begin{aligned} &= T^{-1}dXT + (-T^{-1}dT)X' + X'(T^{-1}dT), \\ &= T^{-1}dXT + [X', T^{-1}dT]. \end{aligned}$$

$$\begin{aligned} d\Theta &:= T^{-1}dT, \\ dX' &= T^{-1}dXT + [X', d\Theta], \\ ddX' &= d(T^{-1})dXT + T^{-1}ddXT + T^{-1}dXdT, \\ &\quad + [dX', d\Theta] + [X', dd\Theta]. \end{aligned}$$

$$\begin{aligned} d(T^{-1})dXT &= (-T^{-1}dT)(T^{-1}dXT), \\ &= -d\Theta(dX' - [X', d\Theta]). \end{aligned}$$

$$\begin{aligned} T^{-1}dXdT &= (T^{-1}dXT)(T^{-1}dT), \\ &= (dX' - [X', d\Theta])d\Theta. \end{aligned}$$

$$\begin{aligned} ddX' &= T^{-1}ddXT, \\ &+ 2[dX', d\Theta], \\ &+ [X', dd\Theta], \\ &+ [d\Theta, [X', d\Theta]]. \end{aligned}$$

$$\begin{aligned} d\Theta/dt &= \Omega, \\ dd\Theta/dt^2 &= d\Omega/dt, \\ dX'/dt &= T^{-1}(dX/dt)T + [X', \Omega], \\ ddX'/dt^2 &= T^{-1}(ddX/dt^2)T, \\ &+ 2[dX'/dt, \Omega], \\ &+ [X', d\Omega/dt], \\ &+ [\Omega, [X', \Omega]]. \end{aligned}$$

Transformed Basis Elements

The transformed basis relations below use the involution rules of Eqs. (7) and (8).

$$\begin{aligned} T(x) &= f(x) + \mathbf{g}(x) \cdot \boldsymbol{\sigma}, \\ T^{-1} &= (f - \mathbf{g} \cdot \boldsymbol{\sigma})/(f^2 - \mathbf{g}^2), \\ dT &= df + d\mathbf{g} \cdot \boldsymbol{\sigma}, \\ \sigma'_n &:= T^{-1}\sigma_n T, \\ \sigma_n &= T\sigma'_n T^{-1}. \end{aligned}$$

Applying the sigma-conjugate and reverse operators gives

$$\begin{aligned} (\sigma'_n)^* &= (T^{-1})^* \sigma_n^* T^*, \\ &= -(T^*)^{-1} \sigma_n T^{*}. \end{aligned}$$

$$\begin{aligned} (\sigma'_n)^R &= T^R \sigma_n^R (T^{-1})^R, \\ &= T^R \sigma_n (T^R)^{-1}. \end{aligned}$$

$$\begin{aligned} (\sigma'_n)^{*R} &= -T^{*R} \sigma_n (T^{-1})^{*R}, \\ &= -T^{-1} \sigma_n T \text{ if } T^{-1} = T^{*R}, \\ &= -\sigma'_n \text{ if } T^{-1} = T^{*R}. \end{aligned}$$

$$\begin{aligned} \boldsymbol{\sigma}' \text{ algebra} &= \boldsymbol{\sigma} \text{ algebra}, \\ (\sigma'_n)^2 &= T^{-1} \sigma_n T T^{-1} \sigma_n T, \\ &= T^{-1} \sigma_n^2 T = 1. \end{aligned}$$

$$\begin{aligned} T^{-1} \sigma_n \sigma_m T &= T^{-1} \sigma_n T T^{-1} \sigma_m T = \sigma'_n \sigma'_m, \\ &= T^{-1} \epsilon_{nmk} i \sigma_k T = \epsilon_{nmk} i \sigma'_k. \end{aligned}$$

$$\begin{aligned} \sigma'_1 \sigma'_2 \sigma'_3 &= T^{-1} \sigma_1 T T^{-1} \sigma_2 T T^{-1} \sigma_3 T, \\ &= T^{-1} \sigma_1 \sigma_2 \sigma_3 T = iT^{-1} T = i. \end{aligned}$$

$$\begin{aligned} d\sigma'_n &= [\sigma'_n, T^{-1}dT], \\ d\boldsymbol{\sigma}' &= [\boldsymbol{\sigma}', T^{-1}dT] \quad \text{vector}, \\ Z &= Z_s + Z_v \cdot \boldsymbol{\sigma}' \quad \text{paravector}, \\ \langle\langle Z \rangle\rangle^2 &= ZZ^{*R}, \\ &= (Z_s + Z_v \cdot \boldsymbol{\sigma}')(Z_s - Z_v \cdot \boldsymbol{\sigma}') \text{ if } T^{-1} = T^{*R}, \\ &= Z_s^2 - Z_v^2 \text{ if } T^{-1} = T^{*R}. \end{aligned}$$

$$\begin{aligned} dZ &= dZ_s + dZ_v \cdot \boldsymbol{\sigma}' + Z_v \cdot d\boldsymbol{\sigma}', \\ &= dZ_s + dZ_v \cdot \boldsymbol{\sigma}' + Z_v \cdot [\boldsymbol{\sigma}', T^{-1}dT]. \end{aligned}$$

$$\begin{aligned} T^{-1}dT &= (f df - g dg)/(f^2 - g^2), \\ &+ (f d\mathbf{g} - df \mathbf{g} - i\mathbf{g} \times d\mathbf{g}) \cdot \boldsymbol{\sigma}/(f^2 - \mathbf{g}^2). \end{aligned}$$

The connection-like quantity

$$\begin{aligned} \Omega &:= T^{-1}(dT/dt), \\ \Omega' &:= T\Omega T^{-1} = (dT/dt)T^{-1}, \\ \text{scalar}(\sigma'_n) &= \text{scalar}(T^{-1}\sigma_n T), \\ &= \text{scalar}(\sigma_n T T^{-1}) \quad \text{cyclic property}, \\ &= \text{scalar}(\sigma_n), \\ &= 0. \end{aligned}$$

$$\begin{aligned} \text{scalar}(\Omega \sigma'_n) &= \text{scalar}(T^{-1}(dT/dt)T^{-1}\sigma_n T), \\ &= \text{scalar}((dT/dt)T^{-1}\sigma_n T T^{-1}) \quad \text{cyclic}, \\ &= \text{scalar}((dT/dt)T^{-1}\sigma_n), \\ &= \text{scalar}(\Omega' \sigma_n). \end{aligned}$$

$$\begin{aligned} k_n + i\omega_n &:= \text{scalar}(2\Omega' \sigma_n), \\ \Omega &= \Omega_s + \frac{1}{2}(k + i\omega) \cdot \boldsymbol{\sigma}', \\ [\boldsymbol{\sigma}', \omega \cdot (i\boldsymbol{\sigma}')] &= -2\omega \times \boldsymbol{\sigma}', \\ [\boldsymbol{\sigma}', \mathbf{k} \cdot \boldsymbol{\sigma}'] &= 2i\mathbf{k} \times \boldsymbol{\sigma}', \\ d\boldsymbol{\sigma}'/dt &= [\boldsymbol{\sigma}', \Omega], \\ &= i(k + i\omega) \times \boldsymbol{\sigma}', \\ &= (-\omega + ik) \times \boldsymbol{\sigma}'. \end{aligned}$$

For $n \neq m$, the mixed products transform as follows.

$$\begin{aligned} Z_{\mathbf{v}} \cdot [\boldsymbol{\sigma}', \Omega] &= (-Z_{\mathbf{v}} \times \omega + iZ_{\mathbf{v}} \times \mathbf{k}) \cdot \boldsymbol{\sigma}', \\ dZ/dt &= dZ_s/dt, \\ &+ (dZ_{\mathbf{v}}/dt - Z_{\mathbf{v}} \times \omega + iZ_{\mathbf{v}} \times \mathbf{k}) \cdot \boldsymbol{\sigma}'. \end{aligned}$$

The same decomposition is useful in non-inertial settings, where the transformed basis carries effective angular-velocity and boost-like terms:

$$\begin{aligned} X &= t + \mathbf{r} \cdot \boldsymbol{\sigma}', \\ dX/dt &= dt/dt, \\ &+ (d\mathbf{r}/dt - \mathbf{r} \times \omega + i\mathbf{r} \times \mathbf{k}) \cdot \boldsymbol{\sigma}'. \end{aligned}$$

The corresponding energy-momentum split is

$$\begin{aligned} mdX/dt &= E + \mathbf{P}' \cdot \boldsymbol{\sigma}', \\ E/m &= dt/dt, \\ \mathbf{P}'/m &= d\mathbf{r}/dt + \mathbf{r} \times (-\omega + i\mathbf{k}). \end{aligned}$$

If $T^{-1} = T^{*\mathbf{R}}$, then the transformed interval keeps the same Minkowski form:

$$\begin{aligned} \langle\langle dX \rangle\rangle^2 &= dt^2 - (d\mathbf{r} - \mathbf{r} \times \omega dt + i\mathbf{r} \times \mathbf{k} dt)^2, \\ &= (1 - (d\mathbf{r}/dt - \mathbf{r} \times \omega + i\mathbf{r} \times \mathbf{k})^2) dt^2. \\ ds' &:= \sqrt{1 - (\mathbf{v} - \mathbf{r} \times \omega + i\mathbf{r} \times \mathbf{k})^2} dt. \end{aligned}$$

Lorentz Boosts

The boost identities in this subsection unpack Eq. (29) and its component consequences.

$$\begin{aligned} \theta &= \operatorname{arctanh}(\beta), \\ \beta &= \tanh(\theta), \\ \gamma &= \cosh(\theta), \\ \gamma\beta &= \sinh(\theta), \\ \mathbf{u} &= (u_1, u_2, u_3), \mathbf{u}^2 = 1, \\ L(\theta, \mathbf{u}) &:= \exp(\tfrac{1}{2}\theta(\mathbf{u} \cdot \boldsymbol{\sigma})), \\ &= \cosh(\tfrac{1}{2}\theta) + \sinh(\tfrac{1}{2}\theta)(\mathbf{u} \cdot \boldsymbol{\sigma}). \\ L^{-1} &= \exp(-\tfrac{1}{2}\theta(\mathbf{u} \cdot \boldsymbol{\sigma})). \end{aligned}$$

The boost inverse satisfies $L^{-1} = L^* = L^{*\mathbf{R}}$. Boosts along the same axis commute:

$$[L(\theta, \mathbf{u}), L(\theta', \mathbf{u})] = 0.$$

Consequently, collinear boosts add by rapidity:

$$\begin{aligned} L(\theta', \mathbf{u}) &= L(\theta_1, \mathbf{u})L(\theta_2, \mathbf{u}), \\ \beta' &= (\beta_1 + \beta_2)/(1 + \beta_1\beta_2), \\ \gamma' &= \gamma_1\gamma_2(1 + \beta_1\beta_2). \end{aligned}$$

Acting on the spacetime paravector $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, the boost gives

$$\begin{aligned} X &= t + \mathbf{r} \cdot \boldsymbol{\sigma}, \\ X' &= L^*XL, \\ &= t' + \mathbf{r}' \cdot \boldsymbol{\sigma}. \end{aligned}$$

$$\begin{aligned} t' &= t \cosh(\theta) - \sinh(\theta)(\mathbf{u} \cdot \mathbf{r}), \\ &= \gamma(t - \beta(\mathbf{u} \cdot \mathbf{r})). \end{aligned}$$

Splitting $\mathbf{r}$ into parts parallel and perpendicular to the boost direction,

$$\begin{aligned} \text{par}(\mathbf{r}) &:= (\mathbf{r} \cdot \mathbf{u})\mathbf{u} \quad \text{parallel to } \mathbf{u}, \\ \text{prp}(\mathbf{r}) &:= \mathbf{r} - \text{par}(\mathbf{r}) \quad \text{perpendicular to } \mathbf{u}, \\ \mathbf{r} &= \text{par}(\mathbf{r}) + \text{prp}(\mathbf{r}), \\ \text{par}(\mathbf{r}') &= \gamma(\text{par}(\mathbf{r}) - \beta t\mathbf{u}), \\ \text{prp}(\mathbf{r}') &= \text{prp}(\mathbf{r}), \\ \mathbf{r}' &= \text{par}(\mathbf{r}') + \text{prp}(\mathbf{r}'), \\ &= \text{prp}(\mathbf{r}) + \gamma(\text{par}(\mathbf{r}) - \beta t\mathbf{u}), \\ &= \mathbf{r} + (\gamma(\mathbf{r} \cdot \mathbf{u} - \beta t) - \mathbf{r} \cdot \mathbf{u})\mathbf{u}. \end{aligned}$$

In the small-velocity limit $\beta \rightarrow 0$,

$$\gamma \Rightarrow 1, t' \Rightarrow t, \mathbf{r}' \Rightarrow \mathbf{r}.$$

The boost preserves the paravector interval:

$$\begin{aligned} \langle\langle X' \rangle\rangle^2 &= X'X'^*, \\ &= (L^*XL)(L^*XL)^*, \\ &= L^*XLL^*X^*L, \\ &= L^*XX^*L, \\ &= \langle\langle X \rangle\rangle^2 L^*L, \\ &= \langle\langle X \rangle\rangle^2. \end{aligned}$$

Spatial Rotations

The rotor identities in this subsection unpack Eq. (28) and its component consequences.

$$\begin{aligned} R(\theta, \mathbf{u}) &:= \exp(\tfrac{1}{2}\theta(\mathbf{u} \cdot i\boldsymbol{\sigma})), \\ &= \cos(\tfrac{1}{2}\theta) + \sin(\tfrac{1}{2}\theta)(\mathbf{u} \cdot i\boldsymbol{\sigma}) \quad \text{rotor.} \\ R^{-1} &= \exp(-\tfrac{1}{2}\theta(\mathbf{u} \cdot i\boldsymbol{\sigma})). \end{aligned}$$

The rotor inverse satisfies $R^{-1} = R^{\mathbf{R}} = R^{*\mathbf{R}}$.

$$\begin{aligned} X' &= R^{\mathbf{R}}XR, \\ &= t' + \mathbf{r}' \cdot \boldsymbol{\sigma}. \end{aligned}$$

$$t' = t,$$

$$\begin{aligned} \mathbf{r}' &= \mathbf{r} \cos(\theta) + (\mathbf{u} \times \mathbf{r}) \sin(\theta), \\ &+ (\mathbf{u} \cdot \mathbf{r})(1 - \cos(\theta))\mathbf{u}, \end{aligned}$$

Rodrigues's formula.

Differentiating the transformed quantities gives

$$\begin{aligned} d(R^{\mathbf{R}}) &= (dR)^{\mathbf{R}} = dR^{\mathbf{R}}, \\ R^{\mathbf{R}}R &= 1, \\ dR^{\mathbf{R}}R + R^{\mathbf{R}}dR &= 0, \\ dR &= -RdR^{\mathbf{R}}R. \end{aligned}$$

The associated bivector-valued angular-velocity element is

$$\begin{aligned}
\Omega &:= -R(dR^R/ds) = (dR/ds)R^R, \\
\Omega^R &= -\Omega, \\
\Omega' &= R^R\Omega R = R(dR/ds), \\
\Omega'^R &= -\Omega', \\
dR/ds &= \Omega R, \\
X' &= R^R X R \Rightarrow R X' = X R, \\
dR X' + R dX' &= dX R + X dR, \\
R dX' &= dX R + X dR - dR X', \\
&= dX R + X dR - dR X'. \\
dX'/ds &= R^R(dX/ds)R + [X', \Omega'], \\
&= R^R(dX/ds + [X, \Omega])R.
\end{aligned}$$

Compositions of Boosts and Rotations

$$\begin{aligned}
T &= L R, \\
T^{-1} &= T^{*R} = R^{*R} L^{*R} = R^R L^*, \\
T &= R L, \\
T^{-1} &= T^{*R} = L^{*R} R^{*R} = L^* R^R, \\
T &= R_1 R_2, \\
T^{-1} &= T^{*R} = R_2^R R_1^R, \\
T &= L_1 L_2, \\
T^{-1} &= T^{*R} = L_2^* L_1^*.
\end{aligned}$$

V. ELECTROMAGNETIC SECTOR

With space-time and boosts in place, the next step is the first major physics payoff: the electromagnetic field, gauge structure, and Lorentz force written in compact STA form.

Fields, potentials, and Maxwell equations

The derivative operator comes first because every later field equation is built from it. Writing time and space derivatives together as a single paravector operator mirrors the earlier unification of temporal and spatial coordinates in dX , and it is this parallelism that makes the electromagnetic formulas compact rather than merely abbreviated.

With the paravector differential operator

$$\begin{aligned}
\partial &:= \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, & \partial^* &= \partial_t - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}, \\
\partial\partial^* &= \partial^*\partial = \partial_t^2 - \boldsymbol{\nabla}^2 =: \square,
\end{aligned} \tag{30}$$

and with $V, \rho, \lambda, S, q \in \mathbb{R}$, $\mathbf{E}, \mathbf{B}, \mathbf{A}, \mathbf{J} \in \mathbb{R}^3$, the electromagnetic field and potential are written as

$$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad \Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}. \tag{31}$$

Here $\Phi \in \mathbb{R} \oplus \mathbb{R}^3$, while F is a purely vector complex paravector in $\{0\} \oplus \mathbb{C}^3 \subset \mathbb{C} \oplus \mathbb{C}^3$. Equation (31) packages electric and magnetic fields into one object F , and scalar and vector potentials into one object Φ . The purpose is not only notational compression: once these quantities are assembled into STA objects, differentiation and transformation act on their full content in a single step.

Boost transformation of the electromagnetic field

With F now defined, the same boost element L from Eq. (29) acts on the field by

$$F' = L^* F L. \tag{32}$$

Equation (32) is the direct field analogue of the event transform $L^* X L$. It shows that F is not an extra object appended to the mechanics, but part of the same algebraic structure: boosts act on F by the same two-sided rule that acts on space-time paravectors, and the usual parallel/perpendicular split of $\mathbf{E}$ and $\mathbf{B}$ follows from the same boost algebra.

The identity

$$\partial\Phi = S + F^*, \quad S := \partial_t V + \boldsymbol{\nabla} \cdot \mathbf{A} \tag{33}$$

encodes

$$\mathbf{E} = -\partial_t \mathbf{A} - \boldsymbol{\nabla} V, \quad \mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}. \tag{34}$$

The compact Maxwell equation is

$$\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \tag{35}$$

Splitting real/imaginary scalar/vector parts of (35) reproduces

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \rho, \quad \boldsymbol{\nabla} \cdot \mathbf{B} = 0, \tag{36}$$

$$\boldsymbol{\nabla} \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad \boldsymbol{\nabla} \times \mathbf{E} = -\partial_t \mathbf{B}. \tag{37}$$

Equations (36) and (37) are standard STA/GA electrodynamics up to sign conventions fixed by (31) and (30) [9, 10, 12]. The advantage of Eq. (35) is that divergence and curl do not need to be introduced as separate primitive operators at this stage. They emerge automatically when the scalar and vector parts of the single STA equation are separated.

Wave equations, gauge, and Lorentz force

Applying ∂^* to (35) gives

$$\square F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma}), \tag{38}$$

whose scalar part yields charge conservation,

$$\partial_t \rho + \boldsymbol{\nabla} \cdot \mathbf{J} = 0. \tag{39}$$

Equation (39) is the continuity equation written in the present notation. For a real scalar gauge function $\lambda \in \mathbb{R}$,

$$\Phi' = \Phi + \partial^* \lambda, \quad F' = F. \tag{40}$$

This gauge transformation shows why the potential Φ is retained even when the field F is physically primary: Φ carries the redundancy, while F carries the gauge-invariant derivative content.

For a particle of real mass $m \in \mathbb{R}$, real charge $q \in \mathbb{R}$, and proper velocity $U \in \mathbb{R} \oplus \mathbb{R}^3$, the proper-force formula is

$$m \frac{dU}{ds} = q \text{real}(FU) = q\gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}), \tag{41}$$

which packages both relativistic power and the Lorentz three-force. The real part in Eq. (41) extracts the physically relevant sector of the product FU . The scalar term gives power transfer and the vector term gives the ordinary Lorentz force, so the standard split between energy change and momentum change is preserved inside one STA formula.

Lagrangian and energy-momentum remarks

The electromagnetic particle Lagrangian is

$$\mathcal{L} = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}), \quad (42)$$

and its Euler–Lagrange equations return $d\mathbf{P}/dt = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. This is the standard relativistic coupling prescription: the free action from Eq. (25) is supplemented by the potential term, and the field-strength form of the force law reappears after variation.

A natural paravector candidate for field energy density and flux is

$$T := \frac{1}{2} F F^R = \frac{1}{2} (\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (43)$$

Editorial note: Eq. (43) is a useful frame-dependent energy-density/Poynting paravector, but by itself it is not the full electromagnetic stress-energy tensor. In STA language the full stress-energy object is more naturally a linear map such as $T(a) = \frac{1}{2} F a F^R$ [9, 11].

Supporting Electromagnetic Derivations

The material collected next writes out the field decomposition, potential reconstruction, source formulas, gauge transformations, force law expansions, and variational identities that underlie Eqs. (30)–(43). On a first reading, this subsection may be skimmed and the forward exposition resumed at Section VI.

Electric and Magnetic Fields

This subsection writes the electromagnetic sector in fully expanded STA form so that the relation between the three-vector fields and the single field paravector is explicit. See Eqs. (30), (31), and (35). Here $V, \rho, S, q \in \mathbb{R}$, $\mathbf{A}, \mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{R} \oplus \mathbb{R}^3$, and $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma} \in \{0\} \oplus \mathbb{C}^3$. Write the spatial derivative vector and the STA derivative operator as

$$\begin{aligned} \partial_{\mathbf{r}} &= \{\partial/\partial r_1, \partial/\partial r_2, \partial/\partial r_3\}, \\ \partial_i &= \partial/\partial r_i, \\ \partial_{\mathbf{r}}^2 &= \partial_1^2 + \partial_2^2 + \partial_3^2, \\ \partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\ &= \partial_t + \sigma_1 \partial_1 + \sigma_2 \partial_2 + \sigma_3 \partial_3. \end{aligned}$$

Collect the electric and magnetic fields into the complex three-vector $\mathcal{F}$ and the field paravector F :

$$\begin{aligned} \mathbf{E} &= \{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3\} && \text{electric field,} \\ \mathbf{B} &= \{\mathbf{B}_1, \mathbf{B}_2, \mathbf{B}_3\} && \text{magnetic field,} \\ \mathcal{F} &= \mathbf{E} + i\mathbf{B} && \text{electromagnetic field,} \\ F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, i = \sigma_1 \sigma_2 \sigma_3, \\ &= \mathcal{F} \cdot \boldsymbol{\sigma}. \end{aligned}$$

$$\begin{aligned} F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= -\mathcal{F}^* \cdot \boldsymbol{\sigma}. \end{aligned}$$

Define the source-density paravector by

$$\begin{aligned} D &= \rho + \mathbf{J} \cdot \boldsymbol{\sigma}, \\ D^* &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Here ρ is the scalar charge density and $\mathbf{J}$ is the current-density three-vector. Maxwell’s equations then read

$$\begin{aligned} \partial F &= D^* && \text{Maxwell's equations,} \\ &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \\ \partial^* F^* &= D. \end{aligned}$$

Expanding ∂F into scalar, vector, and bivector parts gives

$$\begin{aligned} \partial F &= (\partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}})((\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}), \\ &= (\partial_{\mathbf{r}} \cdot \mathbf{E}), \\ &\quad + (\partial_{\mathbf{r}} \cdot \mathbf{B})i, \\ &\quad + (\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &\quad + (\partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E}) \cdot (i\boldsymbol{\sigma}). \end{aligned}$$

Separating the vector components then gives

$$\begin{aligned} \partial_{\mathbf{r}} \times \mathbf{B} &= \mathbf{J} + \partial_t \mathbf{E}, \\ \partial_{\mathbf{r}} \times \mathbf{E} &= -\partial_t \mathbf{B}. \end{aligned}$$

Lorentz Transformation of the Field

Once the electromagnetic field has been assembled into the single paravector F , the boost law from Eq. (32) can be expanded directly in components. For a boost $L = \exp(\frac{1}{2}\theta \mathbf{u} \cdot \boldsymbol{\sigma})$ with $\mathbf{u}^2 = 1$, $\beta = \tanh(\theta)$, and $\gamma = \cosh(\theta)$, the parallel and perpendicular field parts relative to $\mathbf{u}$ obey

$$\begin{aligned} F' &= L^* F L, \\ \text{par}(\mathbf{E}') &= \text{par}(\mathbf{E}), \\ \text{par}(\mathbf{B}') &= \text{par}(\mathbf{B}), \\ \text{prp}(\mathbf{E}') &= \gamma(\text{prp}(\mathbf{E}) + \beta \mathbf{u} \times \mathbf{B}), \\ \text{prp}(\mathbf{B}') &= \gamma(\text{prp}(\mathbf{B}) - \beta \mathbf{u} \times \mathbf{E}). \end{aligned}$$

Equivalently, the transformed fields can be written in the compact form

$$\begin{aligned} \mathbf{E}' &= \text{par}(\mathbf{E}) + \gamma(\text{prp}(\mathbf{E}) + \beta \mathbf{u} \times \mathbf{B}), \\ \mathbf{B}' &= \text{par}(\mathbf{B}) + \gamma(\text{prp}(\mathbf{B}) - \beta \mathbf{u} \times \mathbf{E}), \\ \partial' &= L^* \partial L, \\ \partial' F' &= L^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma})L. \end{aligned}$$

Wave Operator and d’Alembertian

The wave operator is written out here in full so that its factorization into the d’Alembertian and the component field equations can be read directly. See Eq. (30).

$$\begin{aligned} \partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\ \partial^* &= \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}. \end{aligned}$$

The associated d'Alembertian is

$$\begin{aligned}\square &:= \partial_t^2 - \partial_{\mathbf{r}}^2, \\ &= \partial \partial^* = \partial^* \partial, \\ &= \langle\langle \partial \rangle\rangle^2.\end{aligned}$$

$$\partial^2 = \partial_t^2 + \partial_{\mathbf{r}}^2 + 2\partial_t \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma}.$$

This should be distinguished from ∂^2 . The present convention follows Eqs. (31) and (35); different sources may reverse the sign of F , adopt the metric signature $(-, +, +, +)$, or use a different box notation. The corresponding wave equation is

$$\begin{aligned}\square F &= \partial^* \partial F, \\ &= \partial^* (\rho - \mathbf{J} \cdot \boldsymbol{\sigma}).\end{aligned}$$

$$\begin{aligned}\square F^* &= \partial(\rho + \mathbf{J} \cdot \boldsymbol{\sigma}), \\ &= (\square \mathbf{E} + i \square \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= (\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J}) + (-\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho + i(\partial_{\mathbf{r}} \times \mathbf{J})) \cdot \boldsymbol{\sigma}. \\ \partial_t \rho &= -\partial_{\mathbf{r}} \cdot \mathbf{J} \quad \text{conservation of charge.}\end{aligned}$$

In vacuum, where $\rho = 0$ and $\mathbf{J} = 0$, this reduces to

$$\begin{aligned}\square \mathbf{E} &= 0 \Rightarrow \partial_t^2 \mathbf{E} = \partial_{\mathbf{r}}^2 \mathbf{E}, \\ \square \mathbf{B} &= 0 \Rightarrow \partial_t^2 \mathbf{B} = \partial_{\mathbf{r}}^2 \mathbf{B}.\end{aligned}$$

Potential Fields

The potential-based formulation is expanded here because it is the point at which gauge freedom, source reconstruction, and the emergence of the electric and magnetic fields all become transparent at once. See Eqs. (33) and (34). Here $V, S, \rho, \lambda \in \mathbb{R}$, $\mathbf{A}, \mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\Phi \in \mathbb{R} \oplus \mathbb{R}^3$, and $F \in \{0\} \oplus \mathbb{C}^3 \subset \mathbb{C} \oplus \mathbb{C}^3$.

$$\begin{aligned}F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ \Phi &= V + \mathbf{A} \cdot \boldsymbol{\sigma}, \\ \partial \Phi &= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}, \\ &\quad + (\partial_t \mathbf{A} + \partial_{\mathbf{r}} V) \cdot \boldsymbol{\sigma}, \\ &\quad + i(\partial_{\mathbf{r}} \times \mathbf{A}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

$$\begin{aligned}S &:= \text{scalar}(\partial \Phi), \\ &= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}.\end{aligned}$$

The component identities extracted from $\partial \Phi$ are

$$\begin{aligned}\mathbf{E} &= -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, \\ \mathbf{B} &= \partial_{\mathbf{r}} \times \mathbf{A}.\end{aligned}$$

Indeed,

$$\begin{aligned}\partial \Phi &= S + F^*, \\ F^* &= \partial \Phi - S, \\ \text{scalar}(F^*) &= \text{scalar}(\partial \Phi - S) = 0, \\ \text{vector}(F^*) &= \text{vector}(\partial \Phi).\end{aligned}$$

The differential operator acts before the involutions are applied, so

$$(\partial \Phi)^{*R} \neq \Phi^{*R} \partial^{*R}.$$

Equivalently, without introducing S , one may write

$$F^* = (\partial \Phi - (\partial \Phi)^{*R})/2.$$

Sources Derived from the Potential

Using the previously defined potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, scalar part $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$, and identity $F^* = \partial \Phi - S$, one obtains

$$\begin{aligned}F &= \partial^* \Phi^* - S, \\ \partial F &= \partial \partial^* \Phi^* - \partial S, \\ &= \square \Phi^* - \partial S, \\ &= (\square V - \partial_t S) + (-\square \mathbf{A} - \partial_{\mathbf{r}} S) \cdot \boldsymbol{\sigma}, \\ &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.\end{aligned}$$

$$\begin{aligned}\rho &= \square V - \partial_t S, \\ \mathbf{J} &= \square \mathbf{A} + \partial_{\mathbf{r}} S.\end{aligned}$$

In Lorenz gauge, where $S = 0$, these relations reduce to

$$\begin{aligned}\partial_t V &= -\partial_{\mathbf{r}} \cdot \mathbf{A}, \\ \rho &= \square V, \\ \mathbf{J} &= \square \mathbf{A}.\end{aligned}$$

Gauge Transform of the Potential

$$\begin{aligned}\text{scalar}(\partial \Phi) &= S, \\ S &= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}.\end{aligned}$$

Let λ be a scalar gauge function. Then

$$\begin{aligned}\Phi' &= \Phi + \partial^* \lambda, \\ \partial \Phi' &= \partial(\Phi + \partial^* \lambda), \\ &= \partial \Phi + \partial(\partial^* \lambda), \\ &= \partial \Phi + \square \lambda.\end{aligned}$$

Since $\square \lambda$ is again a scalar field, only the scalar part changes:

$$\begin{aligned}\text{scalar}(\partial \Phi') &= S + \square \lambda, \\ \text{vector}(\partial \Phi') &= \text{vector}(\partial \Phi), \\ F' &= F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

Proper Force and Lorentz Dynamics

$$\begin{aligned}mU &= mdX/ds, \\ &= m\gamma + m\gamma \mathbf{v} \cdot \boldsymbol{\sigma}, \\ d/ds &= \gamma d/dt.\end{aligned}$$

The time component gives the relativistic work law,

$$d/dt(m\gamma) = q\mathbf{v} \cdot \mathbf{E}.$$

Here $\mathbf{E}$ denotes the electric field, not the energy operator E , and the magnetic field contributes no work term in the scalar component. The spatial component gives the relativistic Lorentz force law,

$$\begin{aligned}\mathbf{P} &:= m\gamma\mathbf{v}, \\ d\mathbf{P}/dt &= q\mathbf{E} + q\mathbf{v} \times \mathbf{B}.\end{aligned}$$

Together these assemble into

$$d/dt(mU) = q\mathbf{v} \cdot \mathbf{E} + (q\mathbf{E} + q\mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.$$

To expose the paravector content explicitly, expand

$$\begin{aligned}FU/\gamma &= ((\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\ &= \mathbf{E} \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \boldsymbol{\sigma} + (\mathbf{E} \cdot \boldsymbol{\sigma})(\mathbf{v} \cdot \boldsymbol{\sigma}), \\ &\quad + i(\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{v} \cdot \boldsymbol{\sigma}), \\ &= \mathbf{E} \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \boldsymbol{\sigma} + \mathbf{E} \cdot \mathbf{v}, \\ &\quad + i(\mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \mathbf{v} - (\mathbf{B} \times \mathbf{v}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

$$\begin{aligned}\text{real}(FU) &= \frac{1}{2}(FU + UF^R), \\ &= \gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}).\end{aligned}$$

The imaginary part of FU exchanges the electric and magnetic roles and is therefore not the sector used in the physical force law.

$$\begin{aligned}mdU/ds &= q\text{real}(FU), \\ &= \frac{1}{2}q(FU + UF^R).\end{aligned}$$

$$mddX/ds^2 = q\text{real}(FdX/ds) \quad \text{4-force.}$$

Variational Derivation of the Lorentz Force

Starting from the Lagrangian of Eq. (42), together with the potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, proper velocity $U = dX/ds$, and $\gamma = dt/ds$, the action takes the form

$$\begin{aligned}\text{action} &= -m \int ds - q \int \text{scalar}(\Phi^* U) ds, \\ &= \int (-m/\gamma - q\text{scalar}(\Phi^* U/\gamma)) dt, \\ &= \int (-m/\gamma, \\ &\quad - q\text{scalar}((V - \mathbf{A} \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma}))) dt, \\ &= \int (-m/\gamma - qV + q\mathbf{v} \cdot \mathbf{A}) dt.\end{aligned}$$

$$\mathcal{L} = -m/\gamma - q(V - \mathbf{v} \cdot \mathbf{A}) \quad \text{Lagrangian.}$$

The canonical momentum is

$$\begin{aligned}\partial_{\mathbf{v}}\mathcal{L} &= m\gamma\mathbf{v} + q\mathbf{A}, \\ &= \mathbf{P} + q\mathbf{A}.\end{aligned}$$

$$\begin{aligned}d/dt\partial_{\mathbf{v}}\mathcal{L} &= d\mathbf{P}/dt + q\partial_t\mathbf{A} + q(\mathbf{v} \cdot \partial_{\mathbf{r}})\mathbf{A}, \\ \partial_{\mathbf{r}}\mathcal{L} &= -q\partial_{\mathbf{r}}V + q\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}).\end{aligned}$$

Using the vector-calculus identity

$$(\mathbf{v}(t) \cdot \partial_{\mathbf{r}})\mathbf{A}(t, \mathbf{r}) = \partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A}).$$

the Euler–Lagrange equation $d/dt\partial_{\mathbf{v}}\mathcal{L} = \partial_{\mathbf{r}}\mathcal{L}$ yields

$$\begin{aligned}d\mathbf{P}/dt &= -q\partial_{\mathbf{r}}V - q\partial_t\mathbf{A}, \\ &\quad + q(\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \partial_{\mathbf{r}})\mathbf{A}), \\ &= q(-\partial_{\mathbf{r}}V - \partial_t\mathbf{A} + \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A})), \\ &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \\ &= m\text{vector}(dU/ds)/\gamma.\end{aligned}$$

$$\begin{aligned}\mathcal{H} &= (\partial_{\mathbf{v}}\mathcal{L}) \cdot \mathbf{v} - \mathcal{L}, \\ &= m\gamma\mathbf{v} \cdot \mathbf{v} + q\mathbf{v} \cdot \mathbf{A} + m/\gamma.\end{aligned}$$

Adding the explicit potential contribution $qV - q\mathbf{v} \cdot \mathbf{A}$ gives

$$\begin{aligned}&= m\gamma(\mathbf{v}^2 + 1/\gamma^2) + qV, \\ &= m\gamma(\mathbf{v}^2 + 1 - \mathbf{v}^2) + qV, \\ &= m\gamma + qV.\end{aligned}$$

Using the Hamiltonian identity $dH/dt = -\partial_t\mathcal{L}$,

$$\begin{aligned}dH/dt &= -\partial_t\mathcal{L}, \\ d\mathcal{H}/dt &= d(m\gamma)/dt + qdV/dt, \\ &= d(m\gamma)/dt + q\partial_tV + q\mathbf{v} \cdot \partial_{\mathbf{r}}V.\end{aligned}$$

$$\begin{aligned}\partial_tH &= q\partial_tV - q\mathbf{v} \cdot \partial_t\mathbf{A}, \\ d(m\gamma)/dt &= q\mathbf{v} \cdot (-\partial_{\mathbf{r}}V - \partial_t\mathbf{A}), \\ &= q\mathbf{v} \cdot \mathbf{E}.\end{aligned}$$

Electromagnetic Energy-Momentum

$$\begin{aligned}F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ F^R &= (\mathbf{E} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ T &= \frac{1}{2}FF^R, \\ &= \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= T_s + T_{\mathbf{v}} \cdot \boldsymbol{\sigma}. \\ \partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}.\end{aligned}$$

This is a frame-dependent energy-density/Poynting paravector, not by itself the full electromagnetic stress-energy tensor.

Using the generic paravector product rule,

$$\begin{aligned}\partial(ZZ') &= (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}}Z', \\ \partial T &= \frac{1}{2}\partial(FF^R), \\ &= \frac{1}{2}(\partial F)F^R + \frac{1}{2}F(\partial F^R) + \frac{1}{2}[\boldsymbol{\sigma}, F] \cdot \partial_{\mathbf{r}}F^R.\end{aligned}$$

$$\begin{aligned}D &= \rho + \mathbf{J} \cdot \boldsymbol{\sigma} \quad \text{density paravector,} \\ D^* &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}, \\ \partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\ \partial F &= D^*, \\ F^2 &= (\mathbf{E}^2 - \mathbf{B}^2) + 2i(\mathbf{E} \cdot \mathbf{B}) \quad \text{scalar.}\end{aligned}$$

Applying the same product rule to F^2 gives

$$\begin{aligned}\partial(ZZ') &= (\partial Z)Z' + Z(\partial Z') + [\boldsymbol{\sigma}, Z] \cdot \partial_{\mathbf{r}}Z', \\ \partial(F^2) &= (\partial F)F + F(\partial F) + [\boldsymbol{\sigma}, F] \cdot \partial_{\mathbf{r}}F, \\ &= D^*F + FD^* + [\boldsymbol{\sigma}, F] \cdot \partial_{\mathbf{r}}F.\end{aligned}$$

$$\begin{aligned}F &= F_{\mathbf{v}} \cdot \boldsymbol{\sigma}, \\ [\boldsymbol{\sigma}, F] &= 2iF_{\mathbf{v}} \times \boldsymbol{\sigma}, \\ [\boldsymbol{\sigma}, F] \cdot \partial_{\mathbf{r}}F &= -2iF_{\mathbf{v}} \cdot (\partial_{\mathbf{r}} \times F_{\mathbf{v}}), \\ &\quad - 2((\partial_{\mathbf{r}} \cdot F_{\mathbf{v}})F_{\mathbf{v}} - \tfrac{1}{2}\partial_{\mathbf{r}}F_{\mathbf{v}}^2) \cdot \boldsymbol{\sigma}, \\ \square F^2 &= \partial^*(F^2) \quad \text{scalar}.\end{aligned}$$

Supporting Tables

The accompanying tables keep the electromagnetic invariants, source splits, potential formulas, and stress-energy-related component identities next to the corresponding field equations. Tables 15 and 16 record the action of ∂ and $i\partial$ on a generic complex paravector field Z .

Table 15. Components of ∂Z .

real scalar:	$\partial_t a + \boldsymbol{\nabla} \cdot \mathbf{A}$
imaginary scalar:	$\partial_t b + \boldsymbol{\nabla} \cdot \mathbf{B}$
real vector:	$\boldsymbol{\nabla} a + \partial_t \mathbf{A} - \boldsymbol{\nabla} \times \mathbf{B}$
imaginary vector:	$\boldsymbol{\nabla} b + \boldsymbol{\nabla} \times \mathbf{A} + \partial_t \mathbf{B}$

Table 16. Components of $i\partial Z$.

real scalar:	$-\partial_t b - \boldsymbol{\nabla} \cdot \mathbf{B}$
imaginary scalar:	$\partial_t a + \boldsymbol{\nabla} \cdot \mathbf{A}$
real vector:	$-\boldsymbol{\nabla} b - \boldsymbol{\nabla} \times \mathbf{A} - \partial_t \mathbf{B}$
imaginary vector:	$\boldsymbol{\nabla} a + \partial_t \mathbf{A} - \boldsymbol{\nabla} \times \mathbf{B}$

Tables 17 and 18 summarize the electromagnetic invariants and the energy-density/Poynting split built from F .

Table 17. Components of $\frac{1}{2}F^2$.

real scalar:	$\frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2)$
imaginary scalar:	$i\mathbf{E} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
imaginary vector:	$\mathbf{0}$

Table 18. Components of $\frac{1}{2}FF^{\mathbf{R}}$.

scalar:	$\frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)$	energy density
vector:	$\mathbf{E} \times \mathbf{B}$	Poynting vector

Tables 19 and 20 display the component splits of Maxwell's equation and its wave-equation consequence.

Table 19. Components of $\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$.

real scalar:	$\boldsymbol{\nabla} \cdot \mathbf{E} = \rho$
imaginary scalar:	$\boldsymbol{\nabla} \cdot \mathbf{B} = 0$
real vector:	$\partial_t \mathbf{E} - \boldsymbol{\nabla} \times \mathbf{B} = -\mathbf{J}$
imaginary vector:	$\partial_t \mathbf{B} + \boldsymbol{\nabla} \times \mathbf{E} = 0$

Table 20. Components of $\square F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma})$.

real scalar:	$\partial_t \rho + \boldsymbol{\nabla} \cdot \mathbf{J} = 0$
imaginary scalar:	$0 = 0$
real vector:	$\square \mathbf{E} = -\partial_t \mathbf{J} - \boldsymbol{\nabla} \rho$
imaginary vector:	$\square \mathbf{B} = \boldsymbol{\nabla} \times \mathbf{J}$

Table 21 records the component content of the potential identity $\partial\Phi = S + F^*$.

Table 21. Components of $\partial\Phi = S + F^*$.

real scalar:	$\partial_t V + \boldsymbol{\nabla} \cdot \mathbf{A} = S$
imaginary scalar:	$0 = 0$
real vector:	$\partial_t \mathbf{A} + \boldsymbol{\nabla} V = -\mathbf{E}$
imaginary vector:	$\boldsymbol{\nabla} \times \mathbf{A} = \mathbf{B}$

Table 22 is the component form of the product FU/γ used in the Lorentz-force law.

Table 22. Components of FU/γ .

real scalar:	$\mathbf{E} \cdot \mathbf{v}$
imaginary scalar:	$\mathbf{B} \cdot \mathbf{v}$
real vector:	$\mathbf{E} + \mathbf{v} \times \mathbf{B}$
imaginary vector:	$\mathbf{B} + \mathbf{E} \times \mathbf{v}$

Table 23 gives the direct component split of ∂T for the energy-density/Poynting paravector.

Table 23. Components of ∂T .

real scalar:	$\partial_t T_s + \boldsymbol{\nabla} \cdot T_v$
imaginary scalar:	0
real vector:	$\boldsymbol{\nabla} T_s + \partial_t T_v$
imaginary vector:	$\boldsymbol{\nabla} \times T_v$

Tables 24, 25, 26, and 27 collect the remaining differentiated and source-coupled field combinations used in the energy-momentum discussion.

Table 24. Components of ∂F^2 .

real scalar:	$\partial_t(\mathbf{E}^2 - \mathbf{B}^2)$
imaginary scalar:	$\partial_t(2\mathbf{E} \cdot \mathbf{B})$
real vector:	$\boldsymbol{\nabla}(\mathbf{E}^2 - \mathbf{B}^2)$
imaginary vector:	$\boldsymbol{\nabla}(2\mathbf{E} \cdot \mathbf{B})$

Table 25. Components of FD^* .

real scalar:	$-\mathbf{E} \cdot \mathbf{J}$
imaginary scalar:	$-\mathbf{B} \cdot \mathbf{J}$
real vector:	$\rho \mathbf{E} + \mathbf{B} \times \mathbf{J}$
imaginary vector:	$\rho \mathbf{B} - \mathbf{E} \times \mathbf{J}$

Table 26. Components of D^*F .

real scalar:	$-\mathbf{E} \cdot \mathbf{J}$
imaginary scalar:	$-\mathbf{B} \cdot \mathbf{J}$
real vector:	$\rho \mathbf{E} - \mathbf{B} \times \mathbf{J}$
imaginary vector:	$\rho \mathbf{B} + \mathbf{E} \times \mathbf{J}$

Table 27. Components of $D^*F + FD^*$.

real scalar:	$-2\mathbf{E} \cdot \mathbf{J}$
imaginary scalar:	$-2\mathbf{B} \cdot \mathbf{J}$
real vector:	$2\rho \mathbf{E}$
imaginary vector:	$2\rho \mathbf{B}$

VI. RELATIVISTIC WAVE EQUATIONS AND SPINOR DYNAMICS

The relativistic wave-equation sector is a second application of the same algebra: first the scalar wave equation, then minimal coupling and factorization, and only then the Dirac block map and Pauli limit.

Klein–Gordon and frequency splitting

For a complex scalar field $\Psi \in \mathbb{C}$,

$$(\partial\partial^* + m^2)\Psi = 0 \quad (44)$$

is the free Klein–Gordon equation [4, 5]. Define also

$$\Omega := \langle\langle m + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla} \rangle\rangle = \sqrt{m^2 - \boldsymbol{\nabla}^2}, \quad (45)$$

$$(\partial_t \pm i\Omega)\Psi(\pm) = 0, \quad \Psi = \Psi(+) + \Psi(-),$$

which is the positive/negative-frequency factorization of (44). This factorization is useful both conceptually and technically. It separates opposite-frequency sectors of the scalar theory, and it also prepares the reader for the later factorizations in which second-order relativistic equations are rewritten as first-order systems.

Minimal coupling and operator factorization

Minimal coupling is introduced here directly at the operator level. Rather than modifying each component equation separately, one defines energy and momentum operators that already incorporate the potential Φ established in Section V.

Here $\Psi(\pm) \in \mathbb{C}$, while Ω is a scalar differential operator built from $m \in \mathbb{R}$ and $\boldsymbol{\nabla}$. Define also the scalar differential operator E and vector differential operator $\mathbf{P}$ by

$$E := i\partial_t - qV, \quad \mathbf{P} := -i\boldsymbol{\nabla} - q\mathbf{A}. \quad (46)$$

Then

$$(-E^2 + \mathbf{P}^2 + m^2)\Psi = 0 \quad (47)$$

expands to the minimal-coupling Klein–Gordon operator

$$\left(\square + iqS - q^2\langle\langle\Phi\rangle\rangle^2 + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \boldsymbol{\nabla})\right)\Psi = 0. \quad (48)$$

Using the commutators in Eq. (49),

$$[E, P_m] = iqE_m, \quad [P_m, P_n] = iq(\partial_m A_n - \partial_n A_m), \quad (49)$$

one gets the useful factorization

$$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF. \quad (50)$$

Equation (50) is the bridge between the Klein–Gordon and Dirac-type formulas below. The word “bridge” is literal here. The square of a first-order operator built from E and $\mathbf{P}$ produces the second-order relativistic operator together with the field term, which is exactly the structural fact needed before introducing a Dirac-type equation.

Dirac block form, gauge covariance, and Pauli limit

A useful 2×2 block-operator map is

$$W(Z) := \{0, Z; Z^{*\mathbf{R}}, 0\}, \quad W(Z)^2 = \text{diag}(ZZ^{*\mathbf{R}}, Z^{*\mathbf{R}}Z). \quad (51)$$

Here $Z \in \mathbb{C} \oplus \mathbb{C}^3$, and $W(Z)$ lives in the auxiliary block-matrix representation built from paravector entries. The free and coupled Dirac operators act on a two-component block field $\Psi = \{\Psi_1; \Psi_2\}$ in that representation:

$$D_0 := W(i\partial) - m\mathbf{I}, \quad D_0\Psi = 0, \quad (52)$$

$$D := W(i\partial - q\Phi^*) - m\mathbf{I}, \quad D\Psi = 0. \quad (53)$$

Multiplying (53) by $D' := W(i\partial - q\Phi^*) + m\mathbf{I}$ yields

$$D'D\Psi = 0 \Rightarrow \begin{cases} (-E^2 + \mathbf{P}^2 + m^2 - iqF^*)\Psi_1 = 0, \\ (-E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi_2 = 0. \end{cases} \quad (54)$$

The calculation in Eq. (54) explains why the block construction is more than formal packaging. Squaring the first-order system returns Klein–Gordon-type equations with the expected field couplings, so the Dirac form is seen as a controlled factorization of the relativistic wave operator. With the present ordering of $Z = i\partial - q\Phi^* = E - \boldsymbol{\sigma} \cdot \mathbf{P}$ inside $W(Z)$, the upper component carries $-iqF^*$ and the lower component carries $+iqF$. Appendix F records a symbolic check of this sign structure.

Gauge covariance combines the electromagnetic gauge law (40) with the block-field phase transformation

$$\Psi = e^{iq\lambda}\Psi', \quad (55)$$

Equation (55) leaves the form of (53) unchanged.

For the low-energy reduction, using the operators from Eq. (46), one obtains the leading-order Pauli equation for the same block field Ψ ,

$$2mE\Psi = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})\Psi. \quad (56)$$

Solving Eq. (56) for $i\partial_t\Psi$ gives the more familiar Schrödinger-type form

$$i\partial_t\Psi \approx \left(\frac{(-i\boldsymbol{\nabla} - q\mathbf{A})^2}{2m} + qV\right)\Psi - \frac{q}{2m}(\mathbf{B} \cdot \boldsymbol{\sigma})\Psi. \quad (57)$$

This matches the usual Zeeman term [6, 14], provided one remembers that (57) is a leading-order $1/m$ approximation and that later specialization to uniform static $\mathbf{B}$ is used when decoupling spin-up and spin-down sectors. The nonrelativistic theory therefore appears as an approximation internal to the relativistic one, not as an unrelated starting point. The Pauli equation is the leading low-energy shadow of the Dirac block system once the rest-energy scale has been separated and higher-order corrections are neglected.

If $\mathbf{B} = \|\mathbf{B}\|\mathbf{u}$ with $\mathbf{B}, \mathbf{u} \in \mathbb{R}^3$, $\|\mathbf{u}\| = 1$, and $\mathbf{B}$ uniform and constant, decompose

$$\begin{aligned} \Psi &= \Psi(+) + \Psi(-), & \Psi(\pm) &= \Pi(\pm)\Psi, \\ \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \end{aligned} \quad (58)$$

Then Eq. (57) separates into

$$i\partial_t \Psi(\pm) = \left(\frac{\mathbf{P}^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \Psi(\pm). \quad (59)$$

In this special case the projector language acquires an immediate physical meaning: the field direction selects a preferred axis, and the two projector sectors evolve with opposite Zeeman shifts.

Supporting Wave-Equation Derivations

The derivations collected next expand the scalar and spinor wave equations into their intermediate operator identities, explicit component forms, Fourier interpretations, low-energy limits, and spin-sector decompositions. On a first reading, this subsection may be skimmed and the main exposition resumed at Section VII.

Spin-0 Wave Mechanics and Frequency Splitting

Equation (44) is most transparent when read mode by mode. For a plane-wave ansatz

$$\Psi(t, \mathbf{r}) = c e^{-i\omega t + i\mathbf{k} \cdot \mathbf{r}},$$

the Klein–Gordon equation becomes $\omega^2 = \mathbf{k}^2 + m^2$. The operator Ω in Eq. (45) therefore acts on each Fourier mode by multiplication with the positive root

$$\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}.$$

This is the analytic content of the formal factorization

$$\partial_t^2 + \Omega^2 = (\partial_t - i\Omega)(\partial_t + i\Omega).$$

The decomposition $\Psi = \Psi(+) + \Psi(-)$ is thus not merely notation: it separates the positive- and negative-frequency sectors of the scalar theory in the same way that the Fourier transform separates the two energy branches [13].

Root Operators and Fourier Interpretation

The square-root operators in Eq. (45) are easiest to interpret in Fourier space. If

$$\Psi = \sum_n c_n e^{i(\omega_n t + \mathbf{k}_n \cdot \mathbf{r})},$$

then

$$\begin{aligned} \square \Psi &= \sum_n (\mathbf{k}_n^2 - \omega_n^2) c_n e^{i(\omega_n t + \mathbf{k}_n \cdot \mathbf{r})}, \\ \Omega \Psi &= \sum_n \sqrt{m^2 + \mathbf{k}_n^2} c_n e^{i(\omega_n t + \mathbf{k}_n \cdot \mathbf{r})}. \end{aligned}$$

The factorization

$$\square + m^2 = (m - i\sqrt{\square})(m + i\sqrt{\square})$$

is therefore a shorthand for multiplying each Fourier mode by the corresponding dispersion function. The apparently formal root operator is nothing more than the spectral calculus already introduced in Eq. (19), now applied to the wave operator.

Minimal Coupling and Operator Factorization

Once the potential Φ has been introduced, it is more economical to couple at the operator level than at the component level. Acting on a smooth test field Ψ , the operators from Eq. (46) satisfy

$$[E, P_m] \Psi = iqE_m \Psi, \quad [P_m, P_n] \Psi = iq(\partial_m A_n - \partial_n A_m) \Psi.$$

These commutators are the exact place where the electromagnetic field enters the coupled wave mechanics: electric components arise from time-space commutators, while magnetic components arise from the noncommutativity of spatial covariant momenta.

Expanding the squared operators themselves gives

$$\begin{aligned} E^2 \Psi &= (-\partial_t^2 - iq \partial_t V - 2iqV \partial_t + q^2 V^2) \Psi, \\ \mathbf{P}^2 \Psi &= (-\nabla^2 + iq \nabla \cdot \mathbf{A} + 2iq \mathbf{A} \cdot \nabla + q^2 \mathbf{A}^2) \Psi. \end{aligned}$$

Subtracting these expressions and then adding $m^2 \Psi$ is the direct operator expansion behind Eq. (48).

Substituting those commutators into the product of first-order factors yields

$$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) \Psi = (E^2 - \mathbf{P}^2 - iqF) \Psi,$$

while reversing the order gives

$$(E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P}) \Psi = (E^2 - \mathbf{P}^2 - iqF^*) \Psi.$$

This is the structural statement encoded in Eq. (50). The second-order Klein–Gordon operator is recovered from the square of a first-order object, and the field term carried by F is exactly what remains from the noncommuting covariant momenta.

Free Spin- $\frac{1}{2}$ Factorization

Before the electromagnetic field is turned on, the Dirac block map already shows how the scalar and spinor theories are related. With

$$D_0 := W(i\partial) - m\mathbf{I}, \quad D'_0 := W(i\partial) + m\mathbf{I},$$

one has

$$D'_0 D_0 = -(\square + m^2)\mathbf{I}.$$

Thus every free Dirac solution satisfies the Klein–Gordon equation componentwise. Conversely, if Ψ' is a scalar Klein–Gordon solution, then

$$\Psi = \begin{Bmatrix} \Psi' \\ \frac{i}{m} \partial^* \Psi' \end{Bmatrix}$$

solves the free Dirac system. The Dirac equation is therefore a controlled first-order factorization of the same relativistic mass shell rather than a disconnected model.

The Dirac Block System, Gauge Covariance, and Component Form

The block map $W(Z)$ is useful because it converts a paravector Z into a first-order matrix operator whose square is diagonal:

$$W(Z)^2 = \text{diag}(ZZ^{*\mathbf{R}}, Z^{*\mathbf{R}}Z).$$

With

$$Z = i\partial - q\Phi^* = E - \boldsymbol{\sigma} \cdot \mathbf{P},$$

Eq. (53) becomes a first-order system for the two block components Ψ_1 and Ψ_2 :

$$(i\partial - q\Phi^*)\Psi_2 - m\Psi_1 = 0, \quad (i\partial^* - q\Phi)\Psi_1 - m\Psi_2 = 0.$$

At the block-operator level this means

$$W(i\partial - q\Phi^*)^2 = -\text{diag}\left((\partial + iq\Phi^*)(\partial^* + iq\Phi), (\partial^* + iq\Phi)(\partial + iq\Phi^*)\right).$$

Multiplying by the mass-conjugate operator $D' = W(Z) + mI$ gives

$$D'D\Psi = \text{diag}(-E^2 + \mathbf{P}^2 + m^2 - iqF^*, -E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi.$$

The upper and lower components satisfy Klein–Gordon-type equations with opposite field couplings, exactly as recorded in Eq. (54). The sign pattern matters, which is why the caution recorded in Section E is retained explicitly.

Gauge covariance is then immediate. If

$$\Phi' = \Phi + \partial^*\lambda, \quad \Psi = e^{iq\lambda}\Psi',$$

then the phase derivative produced by $i\partial$ is cancelled by the shift in the potential, leaving the form of Eq. (53) unchanged. The direct component actions of ∂ , ∂^* , Φ , Φ^* , iqF , and iqF^* are recorded in Tables 28–33; those tables supply the explicit two-component formulas corresponding to this block system.

Low-Energy Scalar and Spinor Limits

To reach the nonrelativistic scalar theory, remove the fast rest-energy oscillation by writing $\Psi' = e^{-imt}\Psi$. Substituting into the minimally coupled Klein–Gordon equation gives

$$0 = (-E^2 - 2mE + \mathbf{P}^2)\Psi,$$

so at leading order in $1/m$,

$$E\Psi \approx \frac{\mathbf{P}^2}{2m}\Psi.$$

Solving for $i\partial_t\Psi$ yields the Schrödinger form

$$i\partial_t\Psi = \left(\frac{(-i\nabla - q\mathbf{A})^2}{2m} + qV\right)\Psi + O(1/m^2).$$

The scalar low-energy limit is therefore already encoded in the relativistic theory.

For the spin- $\frac{1}{2}$ system, one performs the same rest-phase extraction and then eliminates the small component iteratively. At leading order this yields

$$2mE\Psi = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})\Psi + O(1/m),$$

which is Eq. (56). Solving for $i\partial_t\Psi$ gives Eq. (57): the kinetic Schrödinger operator is supplemented by the Zeeman term

$-\frac{q}{2m}\mathbf{B} \cdot \boldsymbol{\sigma}$ [6, 14]. The derivation is important because it shows that spin coupling is not appended by hand; it survives as the leading field correction left behind by the relativistic first-order theory.

For a uniform static magnetic field $\mathbf{B} = \|\mathbf{B}\|\mathbf{u}$, the projector family from Eq. (16) diagonalizes the Zeeman term:

$$(\mathbf{B} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm\|\mathbf{B}\|\Pi(\pm).$$

The decomposition $\Psi = \Psi(+) + \Psi(-)$ with $\Psi(\pm) = \Pi(\pm)\Psi$ therefore separates the Pauli equation into the two scalar Schrödinger-type equations shown in Eq. (59). It is a clear example of a purely algebraic construction acquiring an immediate physical interpretation.

Supporting Tables

The accompanying operator tables record the component actions used in the wave-equation factorizations and the spinor-field couplings. Tables 28 and 29 record the direct component action of ∂ and ∂^* on a complex paravector test field.

Table 28. Components of $\partial\psi$.

scalar:	$\partial_t\psi_s + \nabla \cdot \psi_v$
vector:	$\partial_t\psi_v + \nabla\psi_s + i(\nabla \times \psi_v)$

Table 29. Components of $\partial^*\psi$.

scalar:	$\partial_t\psi_s - \nabla \cdot \psi_v$
vector:	$\partial_t\psi_v - \nabla\psi_s - i(\nabla \times \psi_v)$

Tables 30 and 31 then show how the potential Φ and its conjugate act componentwise on the same field.

Table 30. Components of $\Phi\psi$.

scalar:	$V\psi_s + \mathbf{A} \cdot \psi_v$
vector:	$V\psi_v + \psi_s\mathbf{A} + i(\mathbf{A} \times \psi_v)$

Table 31. Components of $\Phi^*\psi$.

scalar:	$V\psi_s - \mathbf{A} \cdot \psi_v$
vector:	$V\psi_v - \psi_s\mathbf{A} - i(\mathbf{A} \times \psi_v)$

Tables 32 and 33 complete the operator catalog by recording the field-coupling terms $iqF\psi$ and $iqF^*\psi$.

Table 32. Components of $iqF\psi$.

scalar:	$iq(F_v \cdot \psi_v)$
vector:	$iq\psi_s F_v - q(F_v \times \psi_v)$

Table 33. Components of $iqF^*\psi$.

scalar:	$-iq(F_v^* \cdot \psi_v)$
vector:	$-iq\psi_s F_v^* + q(F_v^* \times \psi_v)$

VII. SYNTHESIS

The main narrative is now complete. The algebraic rules fixed in Section II have been carried through interval geometry, covariant transformations, electromagnetism, and relativistic wave equations without changing the underlying language. That continuity

is the central organizational point of the manuscript: the later physical equations are consequences of the earlier algebraic setup rather than independent formalisms placed side by side.

The appendices that follow are reference and audit material rather than new prerequisites. Appendix A collects the principal formulas in one place, Appendix B gathers supplementary representation links, and Appendices C–E separate verification, historical context, and cautions from the main development.

A. FORMULA COMPENDIUM

Using natural units,

$$\hbar = c = 1,$$

with metric signature

$$(+, -, -, -),$$

the principal formulas developed in the manuscript are collected below.

$dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}$	differential space-time; Eq. (20),
$ds^2 = \langle\langle dX \rangle\rangle^2 = dX dX^*$	space-time metric; Eq. (21),
$\gamma = \frac{dt}{ds} = \frac{1}{\sqrt{1 - \mathbf{v}^2}}$	Lorentz factor; Eq. (22),
$\frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$	proper velocity; Eq. (22),
$m \frac{dX}{ds} = E + \mathbf{P} \cdot \boldsymbol{\sigma}$	proper momentum; Eq. (23),
$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$	electromagnetic field; Eq. (31),
$\partial = \partial_t + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla}$	partial operator; Eq. (30),
$\partial\partial^* = \partial_t^2 - \boldsymbol{\nabla}^2$	d'Alembertian; Eq. (30),
$\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$	potential field; Eq. (31),
$\partial\Phi = S + F^*$	potential gradient; Eq. (33),
$S = \partial_t V + \boldsymbol{\nabla} \cdot \mathbf{A}$	Lorenz scalar; Eq. (33),
$\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$	Maxwell's equations; Eq. (35),
$\partial^* \partial F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma})$	wave equations; Eqs. (30), (38),
$m \frac{d^2 X}{ds^2} = q \operatorname{real} \left(F \frac{dX}{ds} \right)$	Lorentz force; Eqs. (24), (41),
$(\partial\partial^* + m^2)\Psi = 0$	Klein–Gordon equation; Eq. (44),
$\Omega = \langle\langle m + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla} \rangle\rangle = \sqrt{m^2 - \boldsymbol{\nabla}^2}$	frequency operator; Eq. (45),
$(\partial_t \pm i\Omega)\Psi(\pm) = 0, \quad \Psi = \Psi(+) + \Psi(-)$	frequency split; Eq. (45),
$W(Z) := \{0, Z; Z^{*\mathbf{R}}, 0\}$	2×2 matrix; Eq. (51),
$(W(i\partial) - m\mathbf{I})\Psi = 0$	Dirac equation; Eq. (52),
$(-E^2 + \mathbf{P}^2 + m^2)\Psi = 0$	Klein–Gordon with field; Eq. (47),
$E = i\partial_t - qV, \quad \mathbf{P} = -i\boldsymbol{\nabla} - q\mathbf{A}$	energy and momentum operators; Eq. (46),
$(W(i\partial - q\Phi^*) - m\mathbf{I})\Psi = 0$	Dirac with field; Eq. (53),
$2mE\Psi = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})\Psi$	Pauli equation form; Eq. (56),
$\Psi = \Psi(+) + \Psi(-)$	spin-up plus spin-down split; Eq. (58),
$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad \Psi(\pm) = \Pi(\pm)\Psi$	spin projector; Eqs. (16), (58),
$i\partial_t \Psi(\pm) = \left(\frac{\mathbf{P}^2 \mp q\ \mathbf{B}\ }{2m} + qV \right) \Psi(\pm)$	uniform- $\mathbf{B}$ split; Eq. (59),
$X' = T^{-1}XT$	transforms; Eq. (27),
$R = \exp\left(\frac{1}{2}\theta \mathbf{u} \cdot i\boldsymbol{\sigma}\right)$	rotation; Eq. (28),
$L = \exp\left(\frac{1}{2}\theta \mathbf{u} \cdot \boldsymbol{\sigma}\right)$	Lorentz boost; Eq. (29).

The main exposition establishes these formulas in the following order: algebraic primitives in §II, space-time kinematics in §III, transformation theory in §IV, electromagnetism in §V, and relativistic quantum equations in §VI. Supplementary material on representation links, verification, historical context, and cautions is collected in the appendices that follow.

B. REPRESENTATION LINKS AND ADDITIONAL SPIN STRUCTURE

The main development ends with the wave-equation sector. The following appendix gathers spin bookkeeping, matrix realization, and quaternion correspondence material that clarifies those results but is not needed to follow the central progression from algebra to physics.

Spin projectors, ladder operators, and four sectors

The spin projectors along a unit direction $\mathbf{u} \in \mathbb{R}^3$ are again (16). About the z axis, the ladder operators are algebra elements

$$\begin{aligned} a(\pm) &:= \frac{1}{2}(\sigma_1 \pm i\sigma_2), & [\sigma_3, a(\pm)] &= \pm 2a(\pm), \\ a(\pm)^2 &= 0. \end{aligned} \quad (60)$$

Using the projector family (16), the proper-velocity paravector $U \in \mathbb{R} \oplus \mathbb{R}^3$ from Eq. (22), and the block map (51), define for $\epsilon, s \in \{+, -\}$

$$\Lambda(\pm) := \frac{1}{2}(\mathbf{I} \pm W(U)), \quad \Psi(\epsilon, s) := \Lambda(\epsilon)\Pi(s)\Psi. \quad (61)$$

This packages positive/negative energy and spin up/down into one algebraic splitting. Writing the four-sector decomposition explicitly helps organize the later interpretation of states. Once energy sign and spin projection are both represented by projectors, the algebra can distinguish the sectors cleanly without introducing a separate layer of notation for each one.

With the same U and W , one also has

$$(W(U) - \mathbf{I})\Psi = 0. \quad (62)$$

Editorial note: Eq. (62) is not the general Dirac equation. It is better read as an on-shell projector relation for a free state satisfying $W(U)^2 = \mathbf{I}$, after momentum has already been normalized by m .

Pauli matrices and matrix realization

Consistent with Eq. (1), the matrix realization is the usual Pauli basis,

$$\sigma_1 = \{0, 1; 1, 0\}, \quad \sigma_2 = \{0, -i; i, 0\}, \quad \sigma_3 = \{1, 0; 0, -1\}, \quad (63)$$

Specializing the real paravector $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ from Eq. (12) to $a = t$ and $\mathbf{A} = \mathbf{r}$ gives

$$\begin{aligned} X &= t + \mathbf{r} \cdot \boldsymbol{\sigma} = \{t + r_3, r_1 - ir_2; r_1 + ir_2, t - r_3\}, \\ \det X &= t^2 - \mathbf{r}^2. \end{aligned} \quad (64)$$

This matrix realization is included to anchor the paravector formalism to a familiar representation. The determinant reproduces the same Minkowski quadratic form already seen in Eq. (21), so the matrix picture confirms rather than replaces the intrinsic STA interpretation.

Quaternion correspondence

The quaternion identification is

$$\begin{aligned} q_1 &= -\sigma_2\sigma_3, & q_2 &= -\sigma_3\sigma_1, \\ q_3 &= -\sigma_1\sigma_2, & \mathbf{q} &= -i\boldsymbol{\sigma}. \end{aligned} \quad (65)$$

For $a \in \mathbb{R}$ and $\mathbf{b} \in \mathbb{R}^3$, let $Q = a + \mathbf{b} \cdot \mathbf{q} = a - i\mathbf{b} \cdot \boldsymbol{\sigma}$. Then

$$QQ^{\mathbf{r}} = a^2 + \mathbf{b}^2, \quad Q^2 = a^2 - \mathbf{b}^2 - 2ia\mathbf{b} \cdot \boldsymbol{\sigma}. \quad (66)$$

Equation (66) makes the purpose of this correspondence concrete. Quaternion algebra appears here as a restricted sector of the same STA language after multiplication by $-i$, which makes it easier to compare quaternion rotations with the rotor formalism introduced earlier. **A typographical error sometimes appears in the intermediate expansion $Q^2 = a^2 - 2i(\mathbf{b} \cdot \boldsymbol{\sigma}) + (-i)^2(\mathbf{b} \cdot \boldsymbol{\sigma})^2$. The linear term must carry the factor a .**

Supplementary Representation Derivations

The final local development collects the quaternion, generalized-metric, and matrix-realization material that extends the main STA narrative into adjacent representational viewpoints. It remains here because these formulas clarify and compare the earlier constructions rather than serving as prerequisites for them.

Spin via Rotors

The rotor formalism from Eq. (28) is also the cleanest way to express spin-axis changes. For a unit axis $\mathbf{n} \in \mathbb{R}^3$ and angle $\varphi \in \mathbb{R}$, let

$$R = \exp\left(\frac{\varphi}{2} \mathbf{n} \cdot i\boldsymbol{\sigma}\right), \quad R^R R = 1.$$

Then every unit direction $\mathbf{u}$ transforms by

$$R^R(\mathbf{u} \cdot \boldsymbol{\sigma})R = \mathbf{u}' \cdot \boldsymbol{\sigma},$$

with $\mathbf{u}'$ given by the usual Rodrigues formula. If a spinor-like object transforms as $\psi' = R\psi$, then the scalar expectation built from the rotated axis obeys

$$\text{scalar}(\psi'^R(\mathbf{u} \cdot \boldsymbol{\sigma})\psi') = \text{scalar}(\psi^R(\mathbf{u}' \cdot \boldsymbol{\sigma})\psi).$$

In STA language, spin measurements rotate with the apparatus rather than through a separate external matrix rule.

Spin Projectors, Transition Factors, and Ladder Operators

The spin projectors introduced in Eq. (16) and the ladder operators introduced in Eq. (60) admit the following detailed identities:

$$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$$

are idempotents satisfying $\Pi(\pm)^2 = \Pi(\pm)$, $\Pi(+) + \Pi(-) = 1$, and $(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm\Pi(\pm)$. For a second axis $\mathbf{u}'$, the same-side and cross-side products are

$$\begin{aligned} \Pi(\pm)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(\pm) &= \pm(\mathbf{u} \cdot \mathbf{u}')\Pi(\pm), \\ \Pi(\pm)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(\mp) &= \frac{1}{2}(\mathbf{u}'_{\perp} \pm i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma} \Pi(\mp). \end{aligned}$$

If θ is the angle between the two axes, then

$$\frac{1}{2}(1 + \mathbf{u} \cdot \mathbf{u}') = \cos^2 \frac{\theta}{2}, \quad \frac{1}{2}(1 - \mathbf{u} \cdot \mathbf{u}') = \sin^2 \frac{\theta}{2}.$$

These are the projector identities behind the standard spin-transition factors.

About the z -axis one can also introduce the ladder elements

$$\begin{aligned} a(\pm) &:= \frac{1}{2}(\sigma_1 \pm i\sigma_2), \quad [\sigma_3, a(\pm)] = \pm 2a(\pm), \\ a(\pm)^2 &= 0, \end{aligned}$$

with

$$a(+)a(-) = \Pi(+), \quad a(-)a(+) = \Pi(-).$$

They provide the intrinsic STA version of raising and lowering about a chosen axis.

Commutators and the Scope of the Uncertainty Remarks

For unit directions $\mathbf{u}, \mathbf{u}' \in \mathbb{R}^3$,

$$(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) = \mathbf{u} \cdot \mathbf{u}' + i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma},$$

so the commutator is

$$[\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] = 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}.$$

Its norm is controlled by the angle between the axes,

$$\left\| \frac{1}{2}[\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] \right\|^2 = (\mathbf{u} \times \mathbf{u}')^2.$$

This is the algebraic content relevant to spin-axis noncommutativity. It should not be identified with the full Robertson or Robertson–Schrödinger uncertainty relation, which is a statement about state variances rather than bare operator comparison. The caution in Section E is retained for that reason.

Energy and Spin Projectors

Equation (61) introduced the joint energy/spin split. Paired with the spin projectors $\Pi(\pm)$ from Eq. (16) are the energy projectors

$$\Lambda(\pm) = \frac{1}{2}(\mathbf{I} \pm W(U))$$

When U is a unit time-like paravector, Eq. (22) gives $\langle\langle U \rangle\rangle^2 = 1$, hence $W(U)^2 = \mathbf{I}$. The operators $\Lambda(+)$ and $\Lambda(-)$ are therefore idempotents that split the free on-shell block field into positive- and negative-energy sectors, while $\Pi(+)$ and $\Pi(-)$ resolve the same field along a chosen spin axis. On free on-shell states this same construction is often read formally through the replacement $U \sim i\partial/m$, but in the present manuscript that operator shorthand is only heuristic; the actual statement retained in the text is the projector relation itself.

Together they yield the four-sector bookkeeping decomposition

$$\begin{aligned}\Psi &= \Psi(+, +) + \Psi(+, -) + \Psi(-, +) + \Psi(-, -), \\ \Psi(\epsilon, s) &:= \Lambda(\epsilon)\Pi(s)\Psi.\end{aligned}$$

This decomposition is best read as an organizational split once momentum and spin axis have been fixed, not as a substitute for the full interacting Dirac dynamics.

Expectation-Value Shorthand

For normalized spinor-like algebra elements ψ , it is sometimes convenient to borrow Dirac bra-ket notation as a compact shorthand for scalar projections:

$$\begin{aligned}\langle\psi | \psi\rangle &:= \text{scalar}(\psi^R \psi) = 1, \\ \langle\psi | M | \psi\rangle &:= \text{scalar}(\psi^R M \psi).\end{aligned}$$

This notation is mnemonic only; the underlying objects remain STA elements and scalar extraction is still defined algebraically.

Matrix Realization of Paravectors

The matrix representation in Eq. (63) provides a familiar realization of the sigma algebra without changing any of the intrinsic STA statements. Substituting the Pauli matrices into

$$X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$$

gives the 2×2 Hermitian matrix of Eq. (64), whose determinant is

$$\det X = t^2 - \mathbf{r}^2.$$

That determinant is exactly the Minkowski quadratic form already defined intrinsically by XX^* . The matrix picture is therefore confirmatory rather than foundational: it shows that the paravector formalism reproduces the standard model, but the geometric meaning is already present before any matrices are introduced.

Within the same matrix realization, the sigma-conjugate and reverse operators become concrete:

$$M^\dagger := M^{cT}.$$

Here M^c denotes entrywise complex conjugation. In the matrix realization, $M^\dagger$ is the Hermitian adjoint. For the sigma basis one then has

$$\begin{aligned}i^c &= -i, & \sigma_1^c &= \sigma_1, & \sigma_2^c &= -\sigma_2, & \sigma_3^c &= \sigma_3, \\ (\sigma_n \sigma_m)^{cT} &= \sigma_m \sigma_n = -\sigma_n \sigma_m & (n \neq m), \\ \sigma_n^{cT} &= \sigma_n, & (i\mathbf{I})^{cT} &= -i\mathbf{I}, & (i\sigma_n)^{cT} &= -i\sigma_n,\end{aligned}$$

so in the matrix model the reverse operator coincides with the Hermitian adjoint on the sigma basis.

The sigma-conjugate operator and the combined sigma-conjugate/reverse operator also admit direct matrix formulas. With

$$\mathbf{J} := \{0, 1; -1, 0\},$$

one has

$$M^* = \mathbf{J} M^c \mathbf{J}^{-1}, \quad M^{*R} = \mathbf{J} M^T \mathbf{J}^{-1}.$$

These formulas reproduce the sigma-conjugate action $\sigma_n \mapsto -\sigma_n$ and the combined action $(\cdot)^{*R}$ inside the Pauli-matrix realization while leaving the intrinsic algebraic definitions unchanged.

Quaternion Sector

The quaternion basis introduced in Eq. (65),

$$\begin{aligned} q_1 &= -\sigma_2\sigma_3, & q_2 &= -\sigma_3\sigma_1, \\ q_3 &= -\sigma_1\sigma_2, & \mathbf{q} &= -i\boldsymbol{\sigma} \end{aligned}$$

identifies the quaternion units as a restricted bivector sector of the same algebra. For

$$Q = a + \mathbf{b} \cdot \mathbf{q} = a - i\mathbf{b} \cdot \boldsymbol{\sigma},$$

one finds

$$QQ^R = a^2 + \mathbf{b}^2, \quad Q^2 = a^2 - \mathbf{b}^2 - 2ia\mathbf{b} \cdot \boldsymbol{\sigma}.$$

The linear term must carry the factor a . The quaternion correspondence is useful because it lets the reader compare the traditional quaternion rotation formula with the rotor construction from Eq. (28) inside a single notational framework.

Generalized Spacetime Forms and Metric Remarks

For a more general paravector-valued coordinate field,

$$X = f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}, \quad dX = df + d\mathbf{g} \cdot \boldsymbol{\sigma},$$

one still has the intrinsic interval formula

$$\langle\langle dX \rangle\rangle^2 = dX dX^* = df^2 - d\mathbf{g}^2.$$

If df is time-like, then the same algebraic steps as before give

$$\begin{aligned} ds &= \sqrt{df^2 - d\mathbf{g}^2}, & \mathbf{v} &= \frac{d\mathbf{g}}{df}, \\ U &= \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), & \gamma &= \frac{df}{ds}. \end{aligned}$$

Thus the proper-velocity formula survives unchanged under a reparametrization of the scalar and vector coordinates; what changes is only which functions are being used to encode the same paravector interval.

For comparison with matrix notation, one may also write

$$ds^2 = dx^T Q(x) dx$$

for a symmetric matrix $Q(x)$. Standard linear algebra then gives an orthogonal diagonalization $Q = NDN^T$. That statement is useful for coordinate bookkeeping, but it does not eliminate the Lorentzian sign structure already built into $dX dX^*$. Diagonalization reorganizes the quadratic form; it does not convert the indefinite interval into a positive-definite Euclidean norm. That is the limit of the metric comparison and the reason the intrinsic paravector formula remains the primary statement in the manuscript.

C. LITERATURE VERIFICATION AND FORMULA STATUS

The following table records how the principal statements of the manuscript compare with standard literature and with the symbolic checks collected in Appendix F. Its purpose is to distinguish established reformulations from formulas that require qualification, stronger hypotheses, or explicit correction.

STA statement	Literature correspondence	Status	Notes
Eq. (10)	Pauli/GA product split [9, 11]	Verified	Standard scalar+bivector decomposition.
Eq. (21)	Minkowski interval in paravector form [9, 10]	Verified	Equivalent to $dt^2 - d\mathbf{r}^2$.
Eq. (35)	Compact Maxwell equation in STA [9, 10, 12]	Verified	Sign depends on the chosen F and metric conventions.
Eq. (41)	Lorentz force law in STA packaging [9, 12]	Verified	Combines power and force components into one paravector equation.
Eq. (44)	Klein–Gordon equation [4, 5]	Verified	Standard free relativistic scalar wave equation.

Eq. (47)	Minimal-coupling KG equation [12, 13]	Verified	Operator-domain assumptions should be stated when commuting derivatives and fields.
Eq. (53)	Dirac equation in a 2×2 STA block representation [7, 9, 11]	Verified	Representation-dependent, but equivalent in content once the map to spinor components is fixed.
Eq. (54)	Squared Dirac system after STA factorization [7, 9, 11]	Verified	The upper component carries $-iqF^*$ and the lower component carries $+iqF$ with the operator ordering used here.
Eq. (57)	Leading-order Pauli reduction [6, 14]	Verified with caveat	The displayed form is low-energy and omits higher-order electric-field terms.
Eq. (43)	Electromagnetic energy-momentum density	Needs qualification	Useful frame-dependent object, but not the full stress-energy tensor alone.
Robertson inequality variant	Robertson–Schrödinger uncertainty relation [8]	Incorrect	The displayed expression is not the standard state-variance inequality.
Metric diagonalization line $ds^2 = d\mathbf{x}' \cdot d\mathbf{x}'$	Diagonalization of indefinite quadratic form	Incorrect	Diagonalization does not erase the signature signs; one should keep $\pm$ factors explicit.
Eq. (62)	Free on-shell projector condition	Overstated	It should not be labeled as the general Dirac equation.
Quaternion intermediate line	Direct algebra check	Typo	Missing factor of a in the linear term.

D. HISTORICAL COMMENTARY

The physical theories brought together here are historically layered rather than contemporaneous, so the following dates matter for interpretation. The chronology is not decorative: it clarifies that STA is a later unifying language applied to results that originally emerged in different physical settings and historical moments.

1. 1904: Lorentz published moving-frame transformation formulas for electromagnetism [1].
2. 1905: Einstein recast kinematics in terms of the relativity principle and invariant light speed [2].
3. 1908: Minkowski delivered the “Raum und Zeit” lecture; the standard publication appeared in 1909 [3].
4. 1926: Klein and Gordon independently established the relativistic scalar wave equation [4, 5].
5. 1927: Pauli introduced the nonrelativistic spin-magnetic coupling equation [6].
6. 1928: Dirac produced the relativistic spin- $\frac{1}{2}$ equation, leading to the standard matrix formalism later reexpressed in STA language [7].
7. 1966 and after: Hestenes, Baylis, and Doran–Lasenby gave modern geometric-algebra presentations closely aligned with the spirit of this manuscript [9–11].

E. CAUTIONS AND CORRECTIONS

This section gathers the principal formulas that require correction, reinterpretation, or stronger hypotheses than the surrounding exposition might otherwise suggest. It is intended as part of the mathematical record of the paper rather than as commentary external to it.

1. **The displayed “Robertson inequality” is not the standard Robertson or Robertson–Schrödinger uncertainty relation.**
2. **The metric diagonalization line $ds^2 = d\mathbf{x}' \cdot d\mathbf{x}'$ drops the indefinite signature and should not be presented as a Lorentzian metric identity.**
3. **The quaternion square expansion has a missing factor of a in the linear term.**

4. **The line $T = \frac{1}{2}FF^R$ should be read as a frame-conditioned energy-density/Poynting paravector, not the full electromagnetic stress-energy tensor.**
5. **The label “normalized Dirac” for $(W(U) - I)\Psi = 0$ is too strong; it is an on-shell projector statement, not the general Dirac equation.**
6. **An earlier sign variant of Eq. (54) with $+iqF^*$ in the upper component is inconsistent with the operator ordering in $W(i\partial - q\Phi^*)$; the corrected sign is $-iqF^*$.**
7. **Several operator equalities stated in differential form implicitly assume smooth test functions and benign boundary behavior; those hypotheses should be stated whenever commutators of differential operators are used rigorously.**

F. SYMBOLIC VERIFICATION

All hand-worked derivations now appear in the main body, inside the detailed subsection of the topic they support. The appendix is therefore reserved for symbolic checks of the formulas whose sign structure or operator ordering benefits from an independent computational confirmation.

A11. Python Verification of Main Algebraic and Operator Identities

The following checks were carried out symbolically in Python using SymPy and the standard Pauli-matrix realization of the sigma basis. Algebraic identities were reduced directly in matrix form, while differential identities were checked by acting on generic test functions $\psi(t, x, y, z)$ with symbolic potentials and field components. The verification script is `tools/verify_sta_results.py`.

Check	Status	Detail
Eqs. (1)–(2)	Verified	All nine basis products satisfy $\sigma_n \sigma_m = \delta_{nm} + \epsilon_{nmk} i \sigma_k$ in the Pauli realization.
Eq. (10)	Verified	The product $(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma})$ reduces to $\mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$ for symbolic components A_n, B_n .
Eqs. (13)–(15)	Verified	The complex paravector product, the Minkowski-type form $ZZ^{*R} = S^2 - \mathbf{V}^2$, and the Euclidean scalar part $\text{scalar}(ZZ^R) = S ^2 + \ \mathbf{V}\ ^2$ all reduce correctly.
Eqs. (16), (22), (23)	Verified	The idempotents $\Pi(\pm)$ satisfy the projector identities, $UU^{*R} = 1$, and $E^2 - \mathbf{P}^2 = m^2$ follows from $mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$.
Eqs. (33), (34), (35)	Verified	The potential identity gives $S = \partial_t V + \nabla \cdot \mathbf{A}$, $\mathbf{E} = -\partial_t \mathbf{A} - \nabla V$, and $\mathbf{B} = \nabla \times \mathbf{A}$, while ∂F splits into Gauss, Faraday, and Ampère–Maxwell component equations with the manuscript’s signs.
Eqs. (40) and (41)	Verified	The gauge law $\Phi' = \Phi + \partial^* \lambda$ gives $V' = V + \partial_t \lambda$ and $\mathbf{A}' = \mathbf{A} - \nabla \lambda$, and the product FU splits into the expected power term $\gamma \mathbf{v} \cdot \mathbf{E}$ and force term $\gamma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.
Eqs. (48), (49), and (50)	Verified	Acting on a generic test function reproduces the expanded Klein–Gordon operator, the commutators $[E, P_m] = iqE_m$ and $[P_m, P_n] = iq(\partial_m A_n - \partial_n A_m)$, and the factorization $(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF$.
Eqs. (54), (56), and (59)	Verified	Eliminating one Dirac component yields the two squared equations; with the manuscript’s operator ordering, the upper component carries $-iqF^*$ after everything is moved to the left-hand side, while the lower component carries $+iqF$. The exact Pauli identity $(\boldsymbol{\sigma} \cdot \mathbf{P})^2 = \mathbf{P}^2 - q \mathbf{B} \cdot \boldsymbol{\sigma}$ holds on a test function; and for uniform $\mathbf{B} = B_0 \mathbf{u}$ the projectors satisfy $(\mathbf{B} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm B_0 \Pi(\pm)$.
Eq. (66)	Verified	The quaternion correspondence satisfies $QQ^R = a^2 + \mathbf{b}^2$ and $Q^2 = a^2 - \mathbf{b}^2 - 2ia \mathbf{b} \cdot \boldsymbol{\sigma}$; the omitted factor of a in the older intermediate line is indeed an error.

These Python checks do not replace the literature comparison in Section C, but they do separate purely algebraic and operator-theoretic identities from the physically interpretive claims that still require convention-aware reading.

REFERENCES

- [1] H. A. Lorentz, “Electromagnetic phenomena in a system moving with any velocity less than that of light,” *Proceedings of the Royal Netherlands Academy of Arts and Sciences* 6 (1904), 809–831.
- [2] A. Einstein, “Zur Elektrodynamik bewegter Körper,” *Annalen der Physik* 322 (1905), 891–921. DOI: <https://doi.org/10.1002/andp.19053221004>.
- [3] H. Minkowski, “Raum und Zeit,” lecture delivered September 21, 1908; published in *Physikalische Zeitschrift* 10 (1909), 104–111.
- [4] O. Klein, “Quantentheorie und fünfdimensionale Relativitätstheorie,” *Zeitschrift für Physik* 37 (1926), 895–906. DOI: <https://doi.org/10.1007/BF01397481>.
- [5] W. Gordon, “Der Comptoneffekt nach der Schrödingerschen Theorie,” *Zeitschrift für Physik* 40 (1926), 117–133. DOI: <https://doi.org/10.1007/BF01390840>.
- [6] W. Pauli, “Zur Quantenmechanik des magnetischen Elektrons,” *Zeitschrift für Physik* 43 (1927), 601–623. DOI: <https://doi.org/10.1007/BF01397326>.
- [7] P. A. M. Dirac, “The quantum theory of the electron,” *Proceedings of the Royal Society A* 117 (1928), 610–624. DOI: <https://doi.org/10.1098/rspa.1928.0023>.
- [8] H. P. Robertson, “The uncertainty principle,” *Physical Review* 34 (1929), 163–164. DOI: <https://doi.org/10.1103/PhysRev.34.163>.
- [9] D. Hestenes, *Space-Time Algebra*, Gordon and Breach, 1966. See also the 2nd edition, Birkhäuser, 2015, DOI: <https://doi.org/10.1007/978-3-319-18413-5>.
- [10] W. E. Baylis, *Electrodynamics: A Modern Geometric Approach*, Birkhäuser, 1999.
- [11] C. Doran and A. Lasenby, *Geometric Algebra for Physicists*, Cambridge University Press, 2003.
- [12] J. D. Jackson, *Classical Electrodynamics*, 3rd ed., Wiley, 1998.
- [13] W. Greiner, *Relativistic Quantum Mechanics: Wave Equations*, 3rd ed., Springer, 2000.
- [14] J. J. Sakurai and J. Napolitano, *Modern Quantum Mechanics*, 2nd ed., Cambridge University Press, 2017.